Department of Statistics
Savitribai Phule Pune University

ST-606 Research Project

Onion Price Analysis and Forecasting

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Project Guide

Dr. Madhuri G Kulkarni

A REPORT SUBMITTED TO SAVITRIBAI PHULE PUNE UNIVERSITY

TOWARDS PARTIAL FULFILLMENT OF THE DEGREE OF MASTER OF SCIENCE (M.Sc.)

IN STATISTICS

IN THE FACULTY OF SCIENCE AND TECHNOLOGY

SUBMITTED BY

UMA BHOKARE (2306)

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UNDER THE GUIDANCE OF

Dr. MADHURI G KULKARNI

Certificate of the Guide

This is to certify that, the following students of M.Sc. Statistics,

1. Uma Bhokare (2306)
2. Pravin Ahire (2301)

have successfully completed their project titled **Onion Price Analysis and Forecasting** under the guidance of **Dr. Madhuri G Kulkarni** and have submitted this project report on Dec 30, 2024, towards partial fulfillment of requirements for the degree of M.Sc. Statistics in Savitribai Phule Pune University during the academic year 2024-2025.

Dr. Madhuri G Kulkarni
(Project Guide)

Prof. V.K. Gedam
(Head of the Department)

Acknowledgments

We would like to express our sincere gratitude to our project guide, Dr. Madhuri G Kulkarni, for her invaluable guidance and support throughout our project. Her expertise, patience, and encouragement have been instrumental in the successful completion of this project.

Additionally, we are grateful to our Head of Department, Prof. V.K. Gedam, for providing the necessary resources and facilities that enabled us to successfully complete this project.

Finally, we would like to acknowledge all teaching and non-teaching staff for their unwavering support and encouragement throughout our academic journey. We would also like to thank our fellow classmates for their cooperation and support throughout the project.

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Introduction

Price forecasting plays a pivotal role in decision-making across various sectors of the economy, particularly in agriculture, where inherent risks and uncertainties arise from fluctuating yields and low price elasticity of demand. Reliable price forecasts are essential for market participants to make informed production and marketing decisions, and for policymakers to design and implement effective commodity programs. Such forecasts have significant implications for macroeconomic planning, as they influence the stability and performance of commodity markets.

In India, the price volatility of onions is a well-documented challenge, affecting both producers and consumers alike. This unpredictability underscores the need for robust forecasting models to anticipate price trends and mitigate economic impacts. Our project, "Onion Price analysis and Forecasting," seeks to address this challenge by leveraging advanced statistical techniques.

Using historical onion price data from 2013 to 2024, we conducted comprehensive Exploratory Data Analysis (EDA) to identify patterns, trends, and seasonal effects. This was followed by the development of predictive models, including classical time series models such as SARIMA (Seasonal Auto-Regressive Integrated Moving Average) and Holt-Winters.

The findings of this project aim to provide actionable insights for farmers, traders, and policymakers, enabling them to navigate market uncertainties with confidence. By combining rigorous data analysis with diverse modeling approaches, this study contributes to the growing body of research in agricultural price forecasting and highlights the potential of data-driven solutions to address real-world challenges.

Overview of Data

The dataset used in this project consists of 1,44,098 observations and 8 variables, providing comprehensive information about onion market trends across various districts and markets in Maharashtra, India, from 2013 to 2024. Below is a brief description of the dataset's structure and variables:

1. **District.Name:** The name of the district where the market is located (e.g.,Ahmednagar, Nashik).

2. **Market.Name:** The specific market within the district (e.g., Ahmednagar, Chandvad).

3. **Variety:** The type of onion traded, such as Other, Red, Local, White, and Hybrid.

4. **Arrivals..Tonnes.:** The quantity of onion arrivals in the market, measured in tonnes.

5. **Min.Price..Rs..Quintal**.: The minimum price of onions recorded in Rs./Quintal.

6. **Max.Price..Rs..Quintal.:** The maximum price of onions recorded in Rs./Quintal.

7. **Modal.Price..Rs..Quintal.:** The most common (modal) price of onions in Rs./Quintal.

8. **Reported.Date:** The date on which the prices and quantities were reported.

The dataset contains information from 25 districts and 134 unique markets, covering diverse onion varieties. It captures key market trends such as price volatility and seasonal changes, making it a robust foundation for forecasting and predictive modeling. The data has been preprocessed to ensure accuracy and consistency, with missing values handled appropriately during analysis.

Motivation Objectives

Onions are an essential part of daily cooking and play a big role in the economy, especially in India. However, their prices often change a lot because of reasons like weather, how much is grown, storage issues, and problems in the supply chain. These sudden price changes affect everyone from farmers and traders to consumers and policymakers. Predicting onion prices can help reduce these uncertainties, ensuring fair prices for farmers and affordable rates for consumers, while also helping the government make better decisions. That's why we chose this topic of onion price forecasting.

Our study is motivated by three primary objectives:

1.Time Series Analysis of Onion Price.

- To analyze trends and patterns in price over the years.
- To identify seasonal variations and long term trends in price .
- Develop a predictive model to accurately forecast onion prices using historical data

2. **Impact of COVID-19 on Onion Prices**:

- Analyze and compare onion price trends and volatility before and after the COVID-19 pandemic to assess its impact on the supply chain and market dynamics.

Literature Review

1.Seasonal ARIMA to forecast fruits and vegetable agricultural prices

The paper investigates the application of Seasonal ARIMA (SARIMA) models to forecast the prices of fruits and vegetables in the agricultural sector. It highlights the challenges posed by seasonal and non-seasonal variations in price patterns, which significantly impact farmers, traders, and policymakers. By leveraging SARIMA models, the study effectively captures the seasonality inherent in agricultural price data, leading to more accurate and reliable predictions. This approach enables stakeholders to make informed decisions related to production, distribution, and market strategies. The results demonstrate the utility of SARIMA in agricultural price forecasting, emphasizing its role in enhancing food security and reducing economic uncertainties in the agricultural supply chain.

Exploratory Data Analysis

Data Collection: The data obtained from https://agmarknet.gov.in .We have used different datasets of the years which includes the statistical data about onion price and Arrivals in different markets in Maharashtra from 2013 to 2024 .

Data Preprocessing: Before analysis, the collected data underwent preprocessing steps to ensure its quality and consistency. This involved cleaning the data by removing duplicates, handling missing values, and resolving any inconsistencies or errors, reshaping the data, renaming the index, dropping columns, combining data, parsing the dataframe.

Exploratory Data Analysis (EDA): Statistical analysis data Analysis was conducted to present numerical data to uncover patterns, relationships, and trends. It involves applying statistical methods and techniques to draw meaningful conclusions and make informed decisions based on the data. Exploratory Data Analysis was conducted to gain a preliminary understanding of the data.

Visualization of Data:

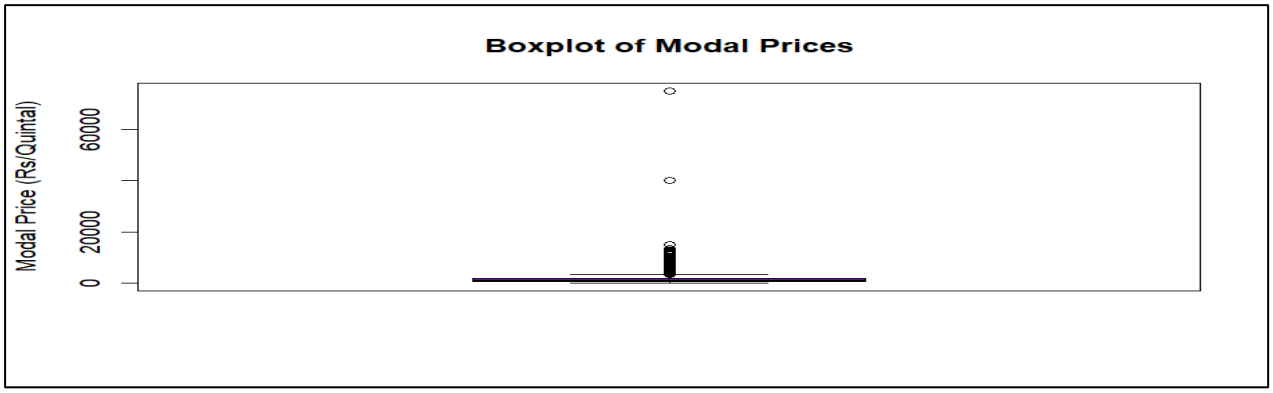
```
> colnames(onion)
[1] "District.Name"             "Market.Name"               "Variety"
[4] "Arrivals..Tonnes."         "Min.Price..Rs..Quintal."   "Max.Price..Rs..Quintal."
[7] "Modal.Price..Rs..Quintal." "Reported.Date"
> head(onion)
  District.Name Market.Name Variety Arrivals..Tonnes. Min.Price..Rs..Quintal.
1    Ahmednagar  Ahmednagar   Other             272.1                     800
2    Ahmednagar  Ahmednagar   Other          1,241.30                    1500
3    Ahmednagar  Ahmednagar     Red          1,269.70                     500
4    Ahmednagar  Ahmednagar     Red          1,559.80                     300
5    Ahmednagar       Akole   Other               5.2                     700
6    Ahmednagar       Akole   Other              85.3                    1000
  Max.Price..Rs..Quintal. Modal.Price..Rs..Quintal. Reported.Date
1                    4750                      2775     24-Aug-13
2                    5500                      4500     28-Sep-13
3                    4500                      2600     26-Sep-13
4                    6000                      3200     07-Nov-13
5                    1751                      1175     24-Jan-13
6                    5200                      4360     05-Sep-13
```

Type of variables in Data:

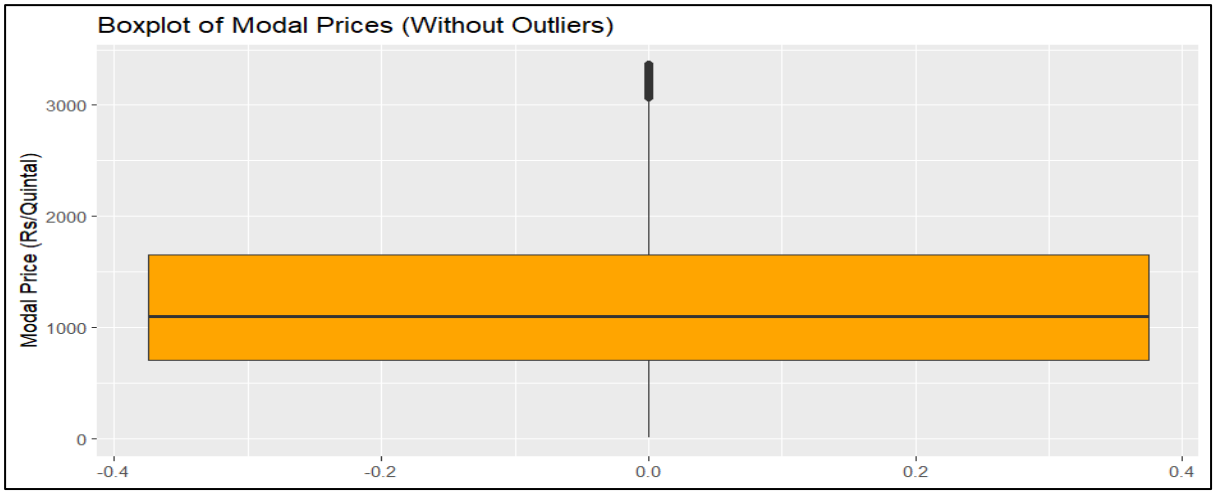
```
> str(onion)
'data.frame':   144098 obs. of  8 variables:
 $ District.Name            : chr  "Ahmednagar" "Ahmednagar" "Ahmednagar" "Ahmednagar" ..
 $ Market.Name              : chr  "Ahmednagar" "Ahmednagar" "Ahmednagar" "Ahmednagar" ..
 $ Variety                  : chr  "Other" "Other" "Red" "Red" ...
 $ Arrivals..Tonnes.        : chr  "272.1" "1,241.30" "1,269.70" "1,559.80" ...
 $ Min.Price..Rs..Quintal.  : int  800 1500 500 300 700 1000 100 3000 200 1000 ...
 $ Max.Price..Rs..Quintal.  : int  4750 5500 4500 6000 1751 5200 2300 5500 1500 2650 ...
 $ Modal.Price..Rs..Quintal.: num  2775 4500 2600 3200 1175 ...
 $ Reported.Date            : chr  "24-Aug-13" "28-Sep-13" "26-Sep-13" "07-Nov-13" ...
> # Get a summary of the dataset
> summary(onion)
 District.Name      Market.Name          Variety          Arrivals..Tonnes.
 Length:144098      Length:144098      Length:144098      Length:144098
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character



 Min.Price..Rs..Quintal. Max.Price..Rs..Quintal. Modal.Price..Rs..Quintal.
 Min.   :    2.0         Min.   :     2          Min.   :    2
 1st Qu.:  300.0         1st Qu.:  1051          1st Qu.:  800
 Median :  500.0         Median :  1600          Median : 1250
 Mean   :  781.4         Mean   :  2055          Mean   : 1582
 3rd Qu.: 1000.0         3rd Qu.:  2600          3rd Qu.: 2000
 Max.   :25000.0         Max.   :373787          Max.   :75000
 Reported.Date
 Length:144098
 Class :character
 Mode  :character
```

BoxPlot of Numerical Variables

Boxplot of Modal Prices

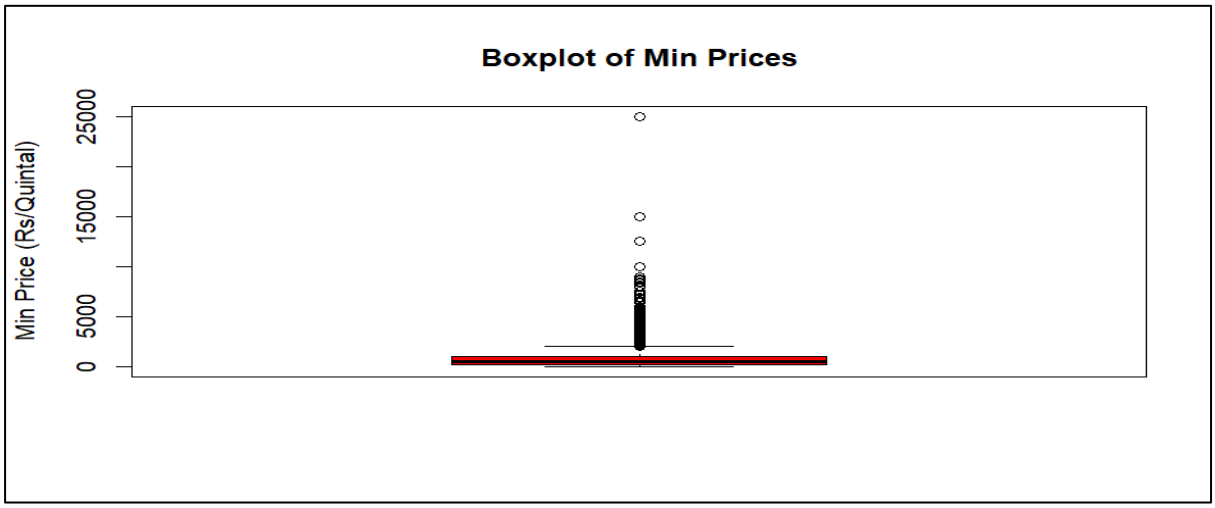


Most modal prices are concentrated below 2,000 Rs/Quintal, as shown by the narrow box and short whiskers. However, some extreme outliers exceed 60,000 Rs/Quintal, creating a highly skewed distribution. While most markets report reasonable prices, a few markets show unusually high values. Further analysis is needed to understand the reasons behind these outliers, such as regional or seasonal factors



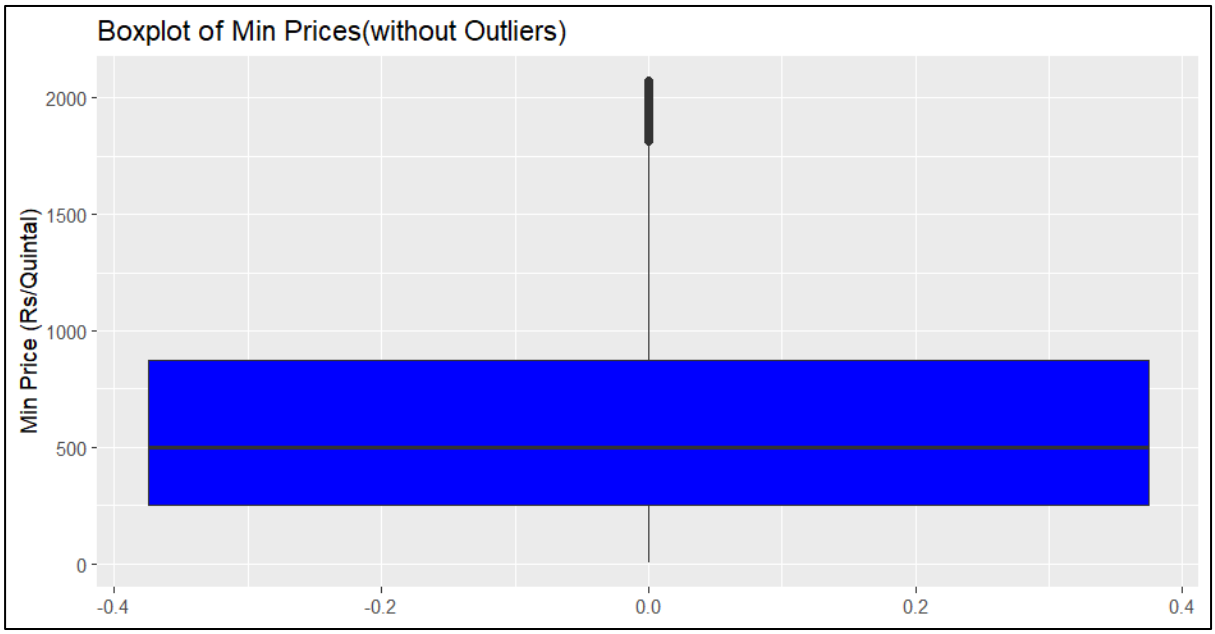
The modal prices for onions are mostly concentrated between 1,000 and 2,000 Rs/Quintal.
A few exceptionally high values (above 3,000 Rs/Quintal) are present as outliers.
The distribution shows a mild right skew, indicating that while most prices are within the normal range, some markets report significantly higher modal prices.

BoxPlot of Minimum Prices



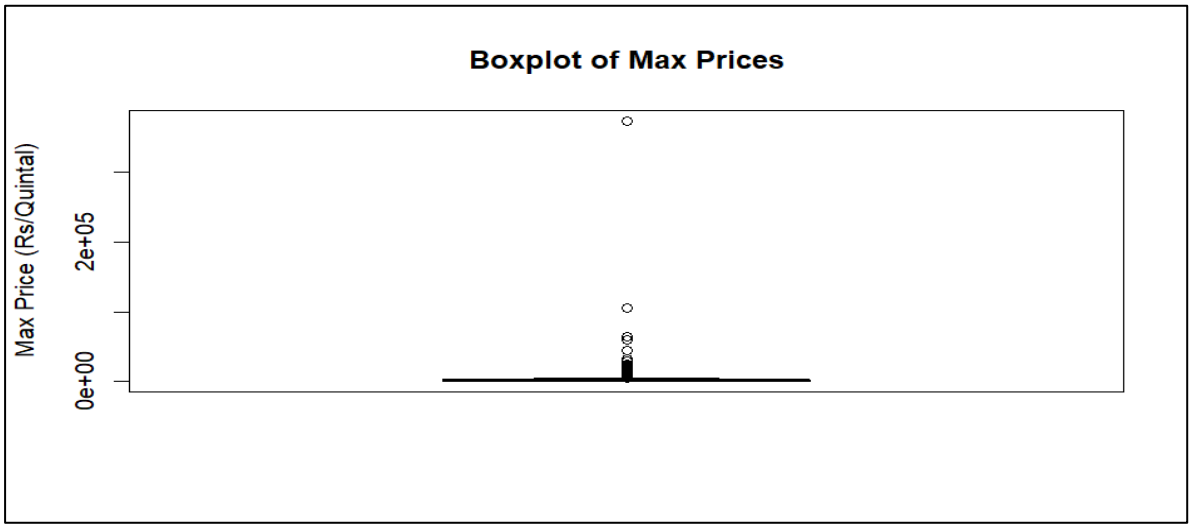
The boxplot shows the distribution of minimum prices (Rs/Quintal). The majority of prices are concentrated below approximately 2,500 Rs/Quintal (indicated by the box), with a median around 1,000 Rs/Quintal. Several outliers are present, indicating a few instances with significantly higher minimum prices, reaching up to 25,000 Rs/Quintal.

Minimum price of onion is 100 Rs/quintal Onion prices fluctuate due to factors like harvest seasons (Kharif: July-October, Rabi: October-March), festivals (Diwali, Holi), supply and demand, and government interventions. Lower prices can occur during harvest seasons, while festivals and seasonal demand can increase prices.

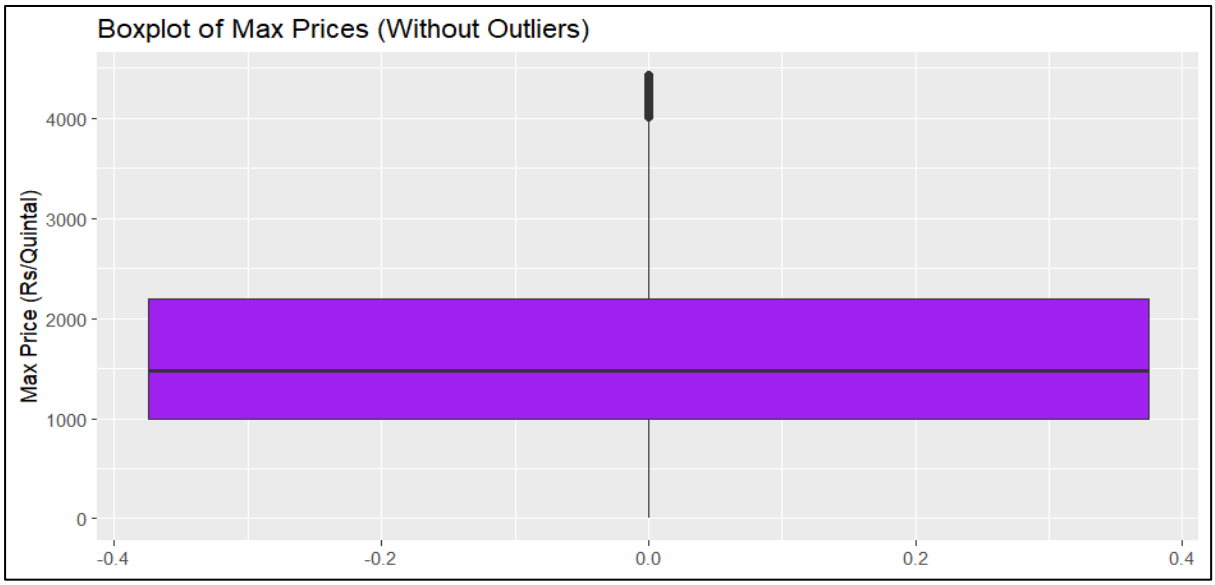


The boxplot displays the distribution of minimum prices (Rs/Quintal) after outliers have been removed. The prices are primarily concentrated within a range from approximately 250 to 900 Rs/Quintal (indicated by the blue box), with a median around 500 Rs/Quintal. The whiskers extend to roughly 0 and 2100 with no outliers present beyond these values.

BoxPlot of Maximum Prices

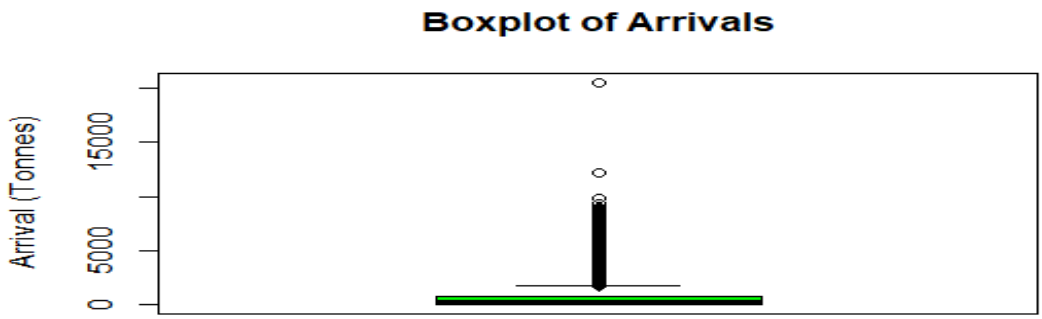


The boxplot shows the distribution of maximum prices (Rs/Quintal). The majority of prices are concentrated near zero (as indicated by the compressed box), with a median very close to zero. There are several high outliers, indicating some instances with significantly higher maximum prices, with one outlier reaching approximately 100,000 Rs/Quintal.

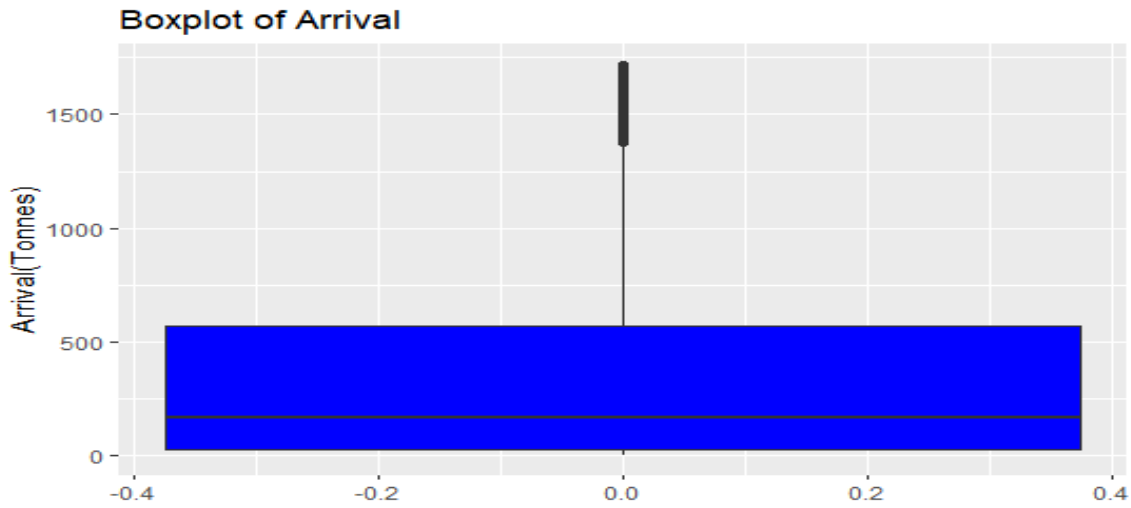


The boxplot displays the distribution of maximum prices (Rs/Quintal) after outliers have been removed. The prices are primarily concentrated within a range from approximately 0 to 1000 Rs/Quintal (indicated by the purple box), with a median around 500 Rs/Quintal. The whiskers extend to roughly 0 and 2000 with no outliers present beyond these values.

BoxPlot of Arrivals

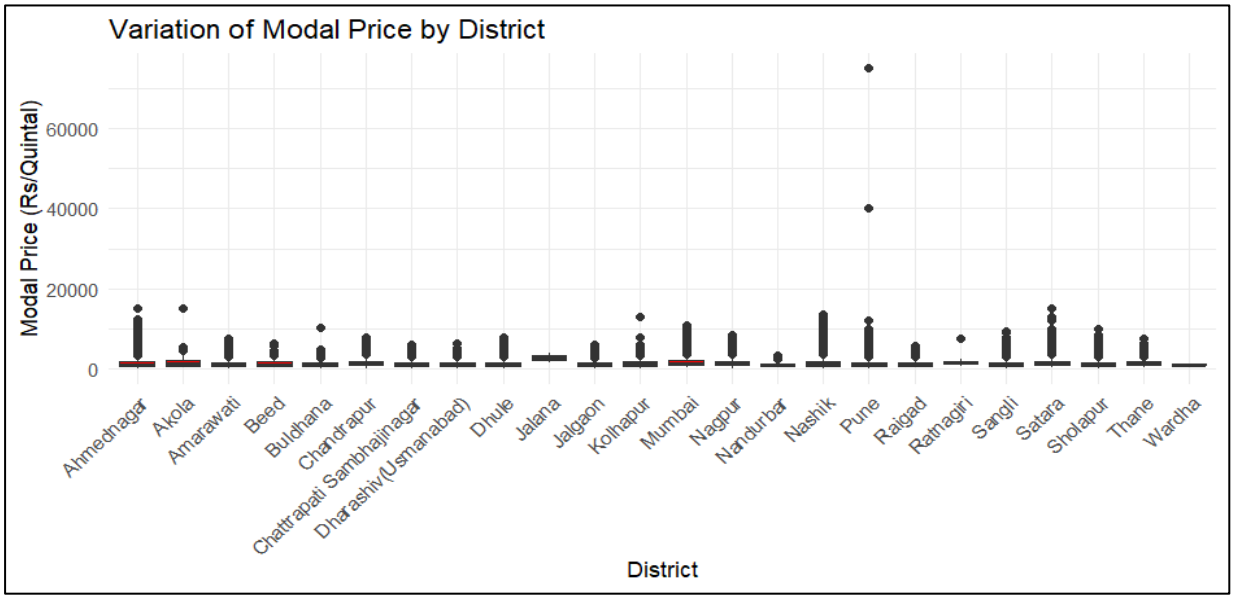


The box plot displays the distribution of arrival prices. The median price is around Rs. 5000 per quintal, indicated by the line within the box. The box itself represents the interquartile range (IQR), showing that 50% of the prices fall between approximately Rs. 4000 and Rs. 6000 per quintal. The whiskers extend to the minimum and maximum values within 1.5 times the IQR, with potential outliers indicated by individual dots beyond the whiskers.



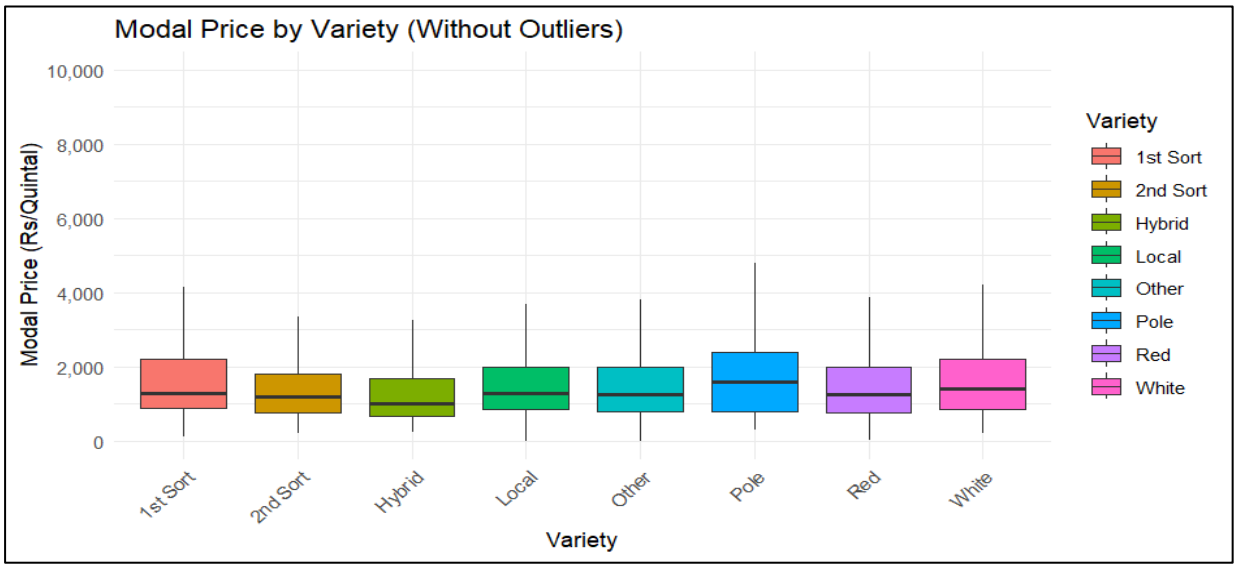
The boxplot of arrival shows a median arrival of around 200 tonnes. The middle 50% of arrivals are clustered within a narrow range, indicating relatively consistent arrival quantities. However, the presence of outliers suggests that there are instances of significantly higher arrival quantities.

Boxplot of Variation of Modal Price by District



The box plot illustrates the variation of modal prices across districts. While districts like Jalgaon and Nashik exhibit higher modal prices, others like Akola and Amravati show lower values. This disparity is likely influenced by factors such as local demand, supply chain efficiency, and market infrastructure. Notably, the presence of outliers in some districts suggests unusual price fluctuations or unique market dynamics that deviate from the general trend.

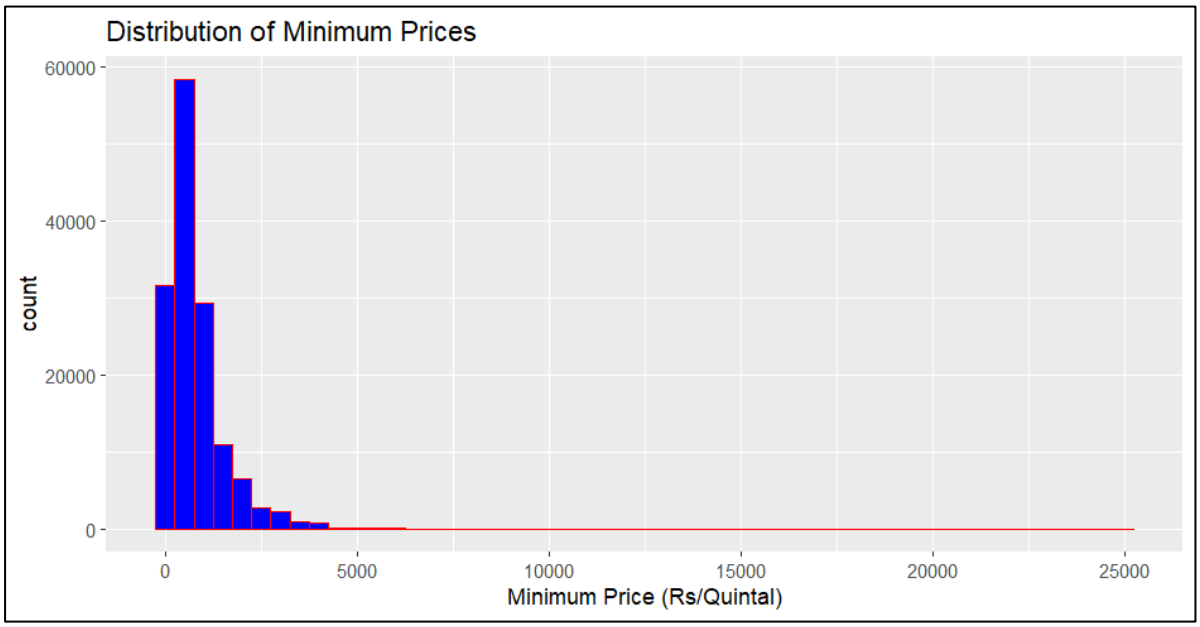
Boxplot of Modal Price by Variety



The box plot displays the distribution of modal prices across different varieties of agricultural produce. Varieties like "1st Sort" and "Red" tend to have higher modal prices, while "2nd Sort" and "Hybrid" varieties exhibit lower values. This variation is likely influenced by factors such as quality, demand, and market preferences. The plot also reveals potential

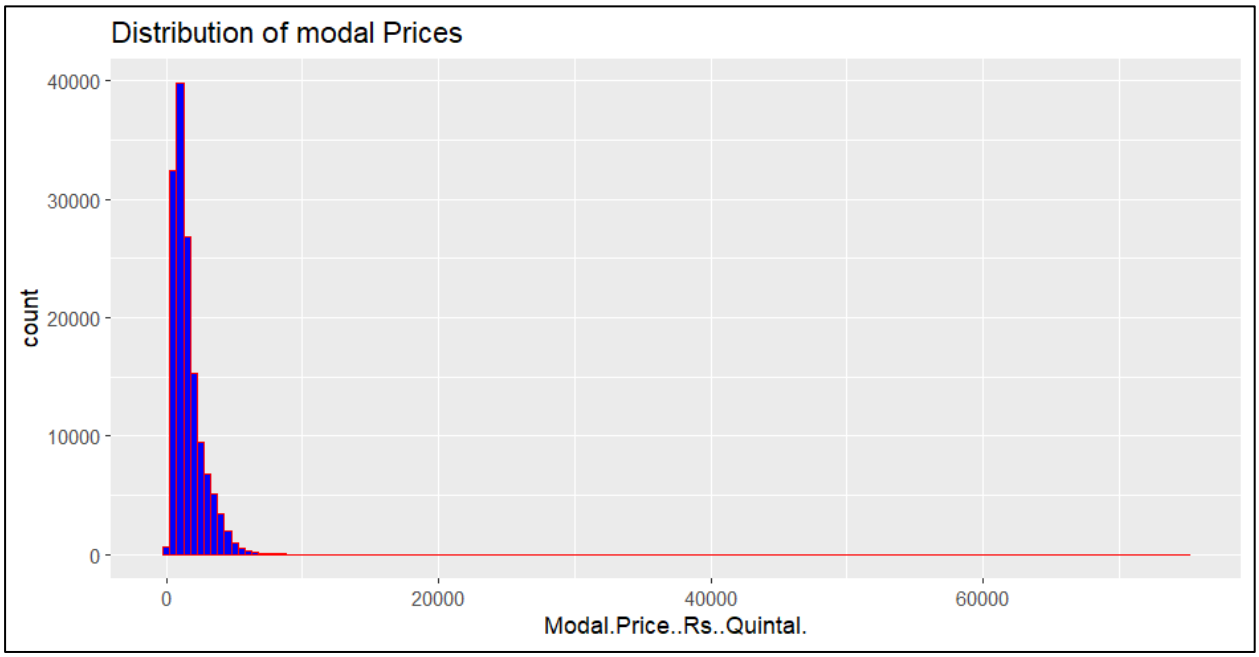
outliers in some varieties, suggesting unusual price fluctuations or unique market dynamics that deviate from the general trend.

Histogram for Minimum Price



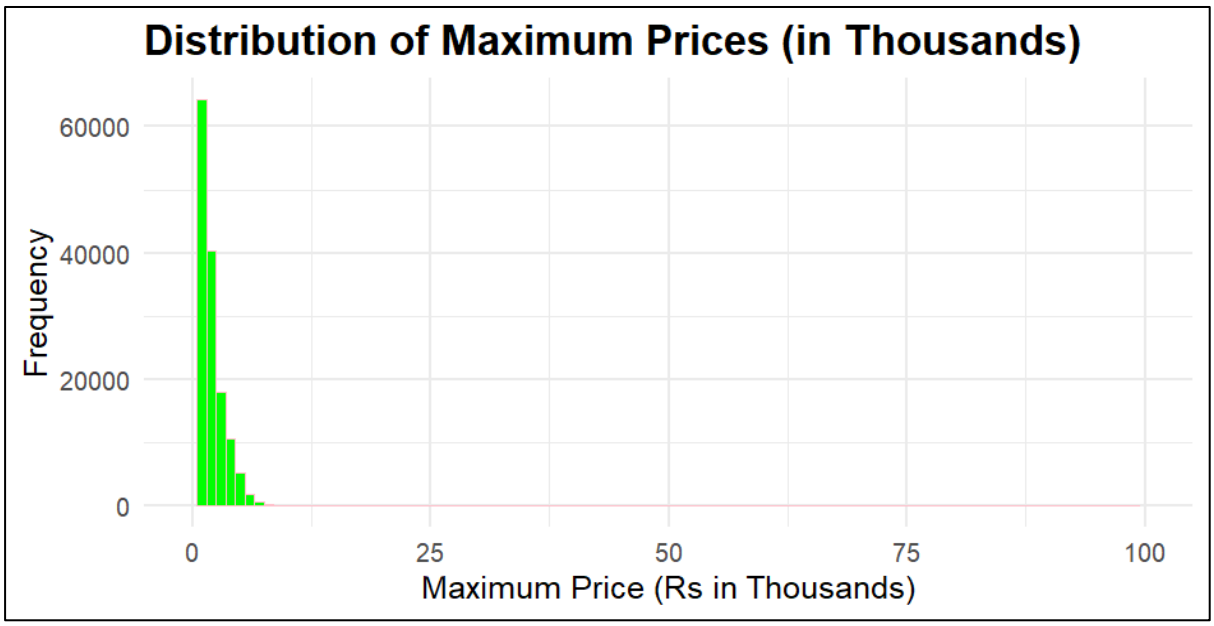
The histogram shows the distribution of minimum prices. Most prices are concentrated between 0 and 5,000 Rs/Quintal, with a peak around 2,000. The distribution is right-skewed, meaning higher prices are less common

Histogram for Modal Prices



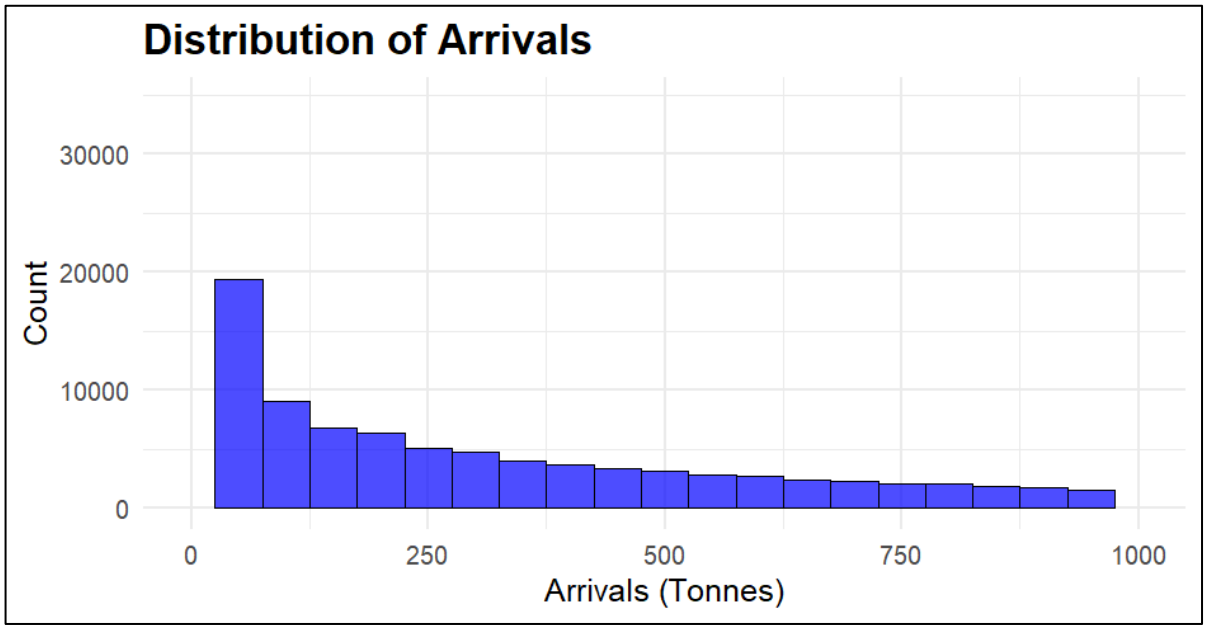
The histogram displays the distribution of modal prices. Most prices are concentrated between 0 and 5,000 Rs/Quintal, with a peak near 2,000. The distribution is skewed right, meaning higher prices are less frequency

Histogram for Maximum Prices



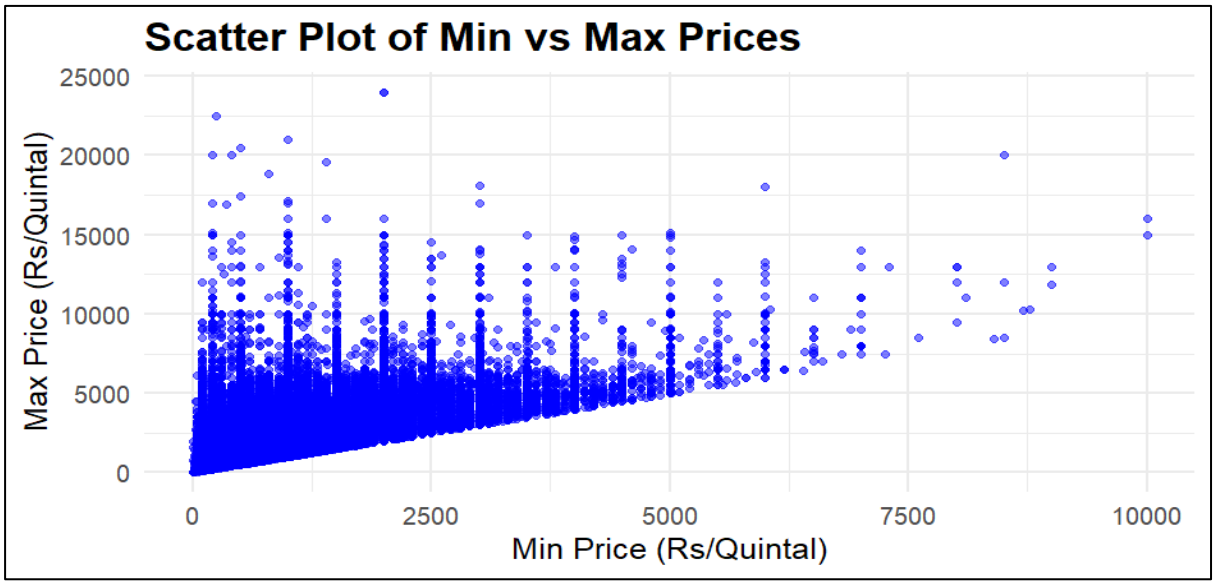
The histogram shows the distribution of maximum prices. Most prices are concentrated between 0 and 25,000 Rs, with a peak near 0. The distribution is skewed right, meaning higher prices are less frequent.

Histrogram for Total Arrivals



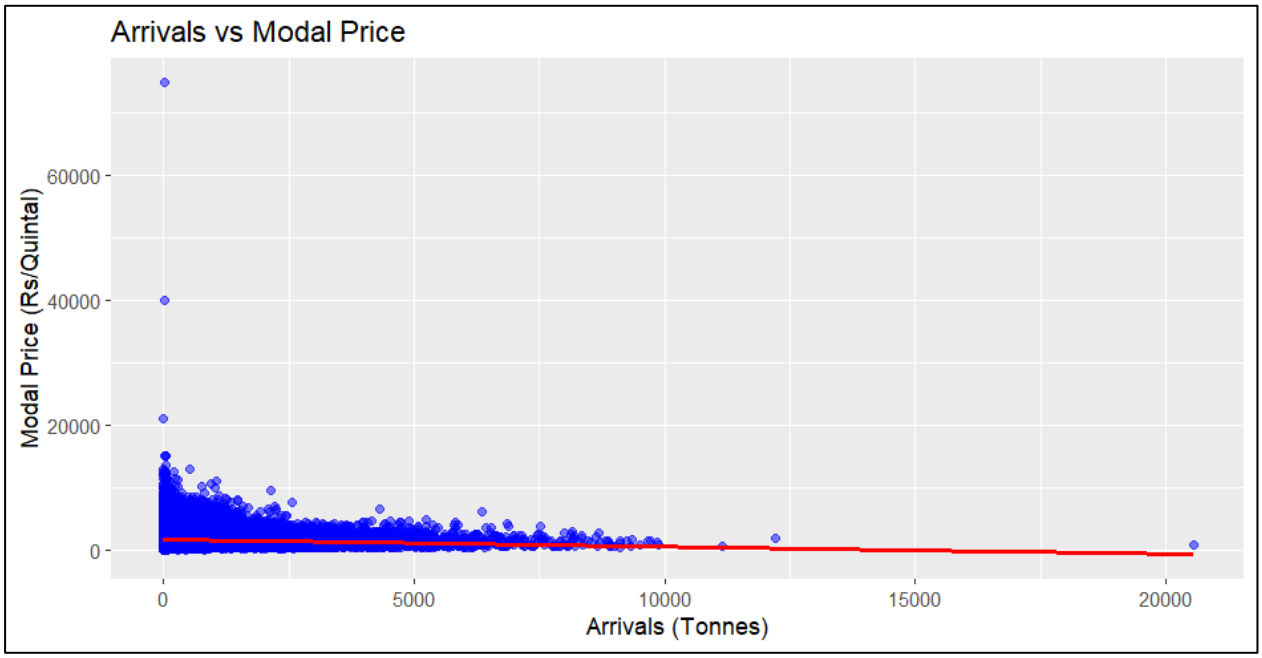
The histogram shows the distribution of arrivals in tonnes. Most arrivals are concentrated between 0 and 250 tonnes, with a peak near 0. The distribution is skewed right, meaning larger arrivals are less frequent.

Scatter Plot between Min Price and Max Price



The scatter plot illustrates the relationship between minimum and maximum prices. There's a clear positive correlation, meaning that as the minimum price increases, the maximum price tends to increase as well. The data points cluster more densely in the lower range of minimum prices, suggesting that a larger proportion of observations fall within this range.

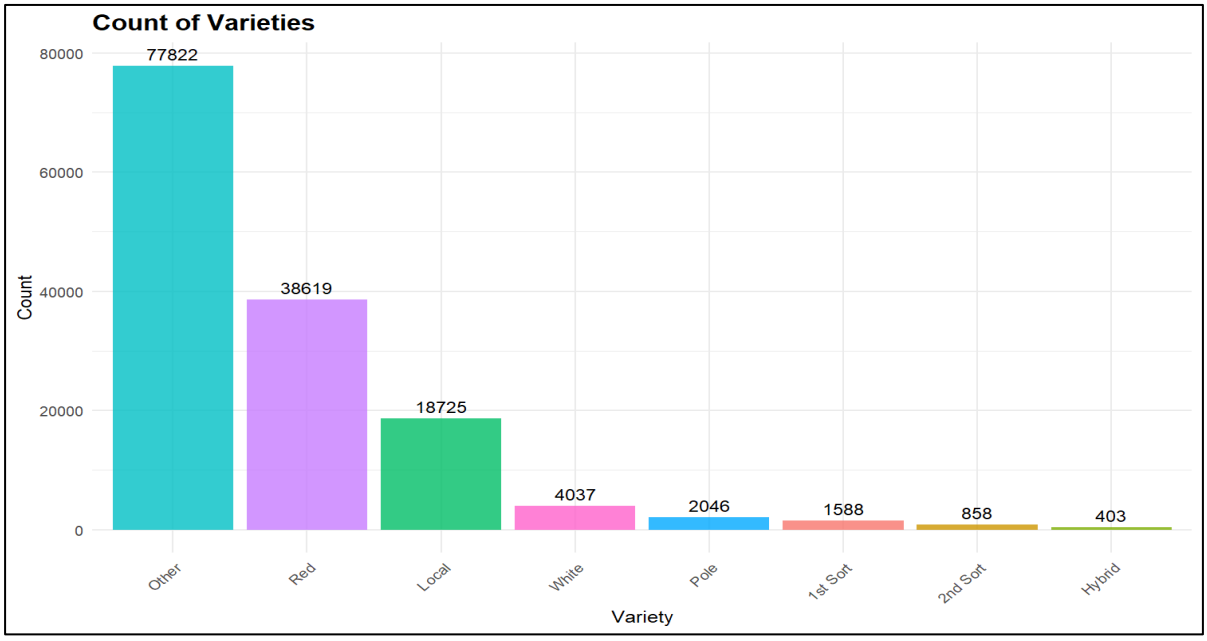
Scatter plot: Arrivals vs Modal Price



The scatter plot examines the relationship between arrivals (likely in tonnes) and modal price. There seems to be a slight negative correlation, suggesting that as the quantity of arrivals increases, the modal price tends to decrease slightly. However, the relationship is not very strong, and there is a lot of variability in the data.

Bar Plots

Bar plot for Variety

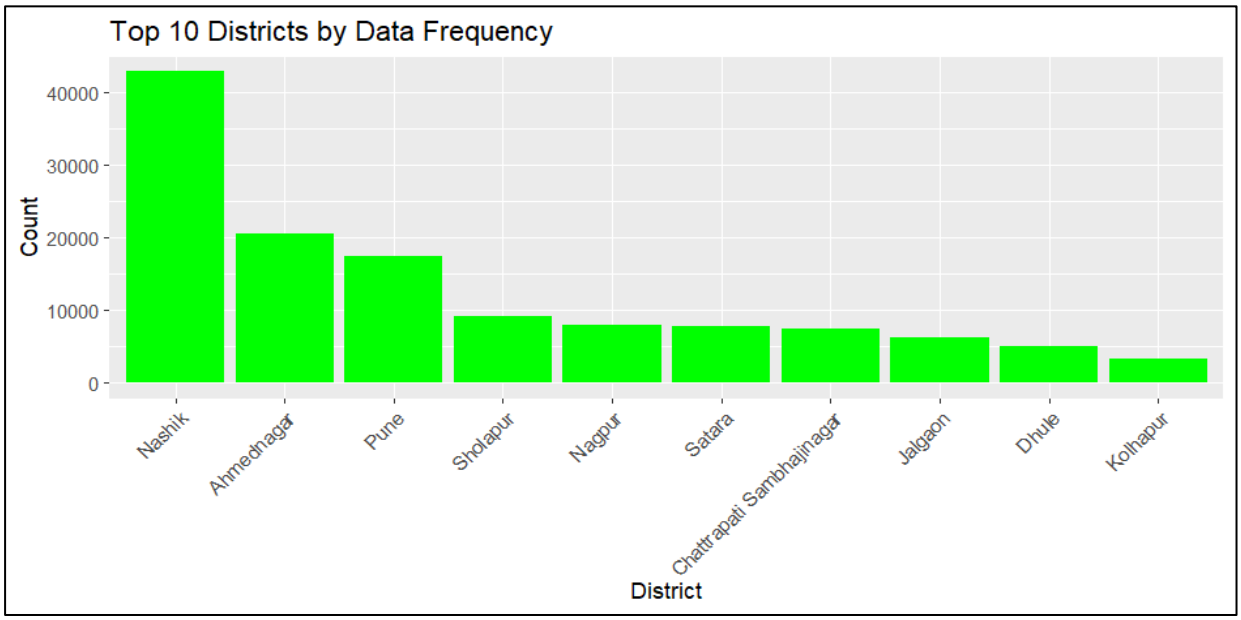


The bar chart illustrates the distribution of different varieties of a product. "Other" is the most common variety, followed by "Red." The remaining varieties, such as "Local," "White," and "Pole," are significantly less frequent compared to the top two categories.

Bar plot for average modal price by Variety

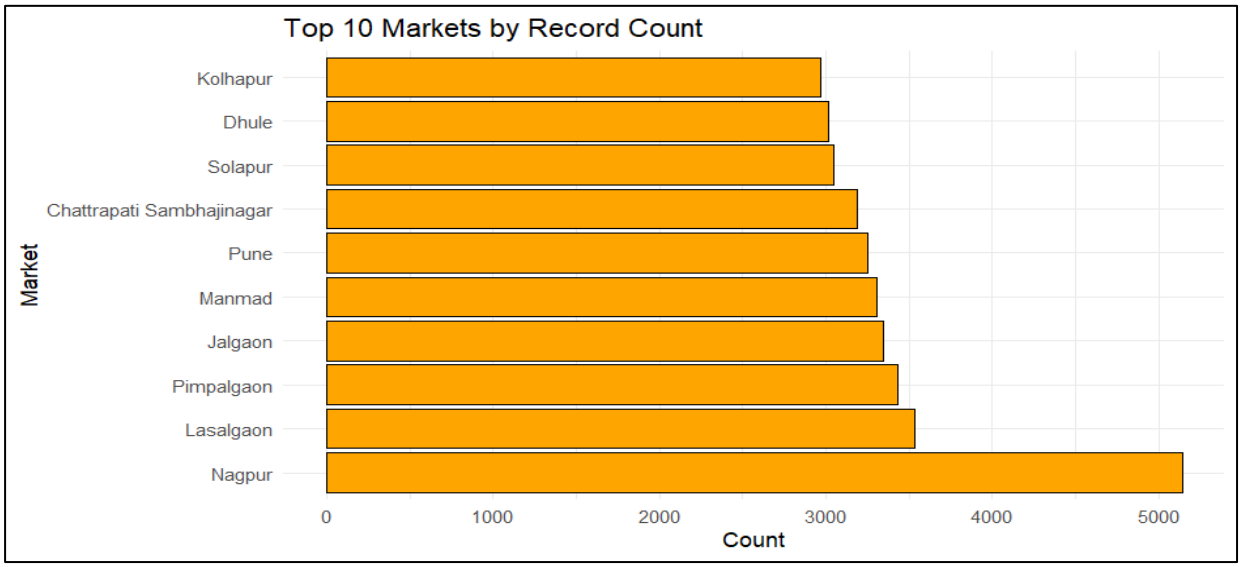
The bar chart compares the average modal prices of different varieties of a product. "Pole" and "1st Sort" have the highest average prices, while "2nd Sort" and "Hybrid" have the lowest. These price differences likely reflect variations in quality, market demand, or production costs associated with each variety.

Bar plot of top 10 districts with the most data



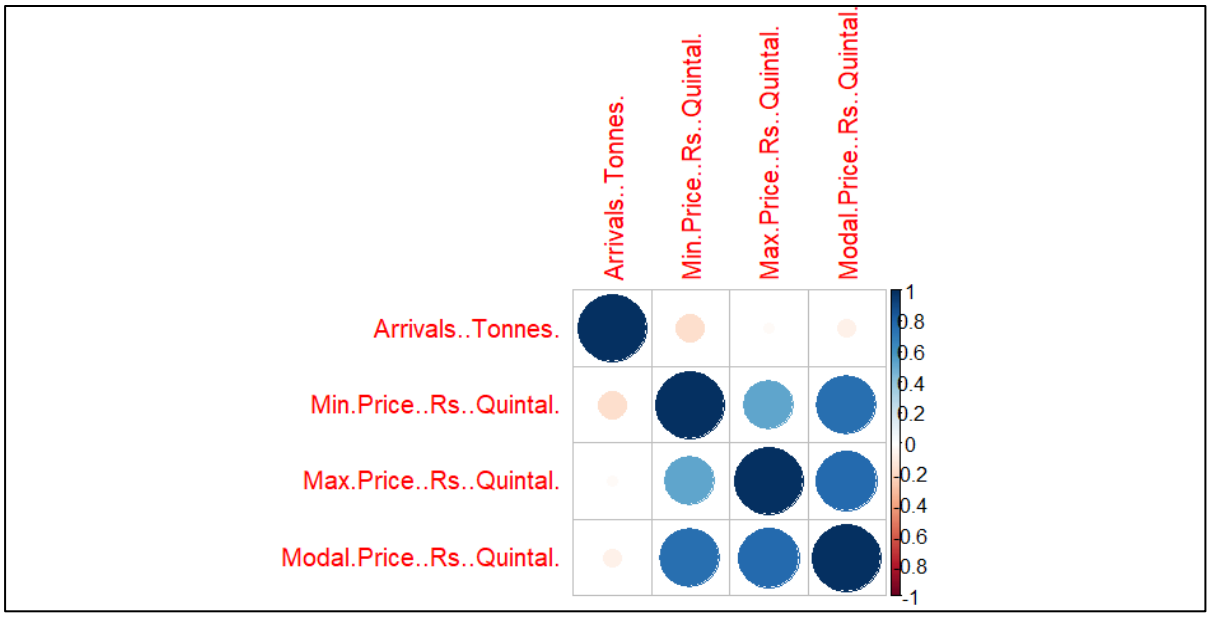
The bar chart presents the frequency of data collected from the top 10 districts. Nashik has the highest data frequency, followed by Ahmednagar and Pune. The data frequency decreases steadily from Nashik to Kolhapur, which has the lowest frequency. This suggests that the data collection efforts were more concentrated in Nashik and its surrounding districts, with decreasing coverage towards the districts with lower frequencies.

Top 10 markets with the most data

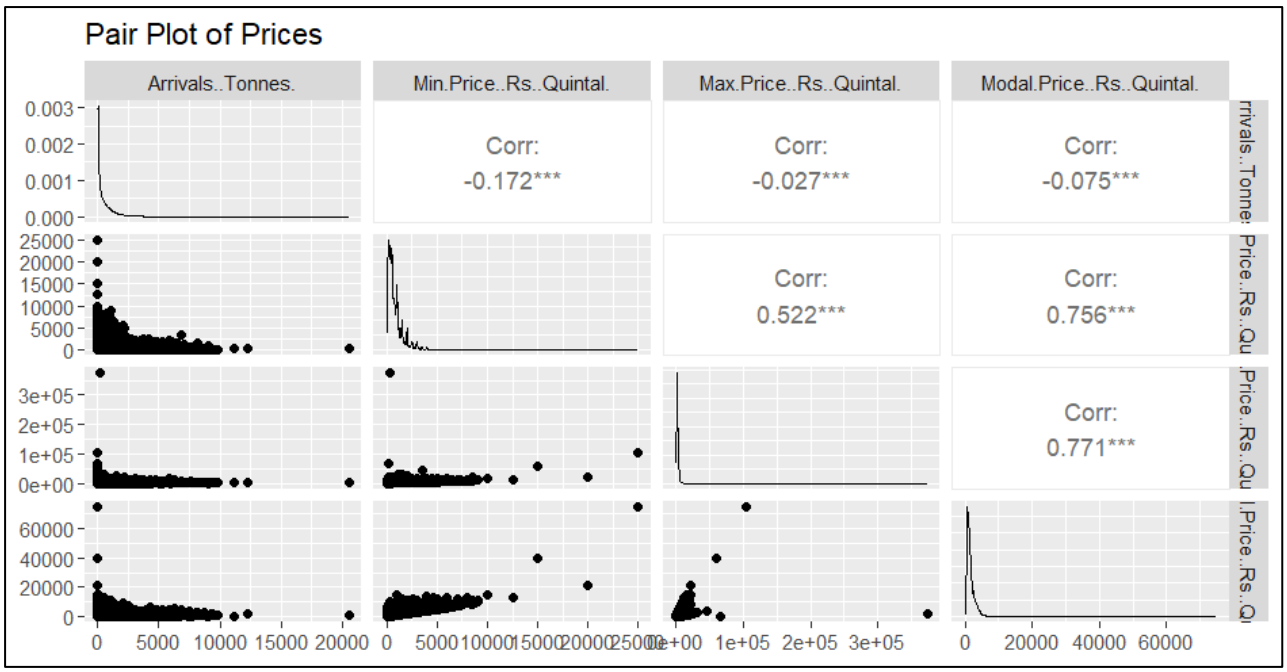


The bar chart presents the top 10 markets by record count. Nagpur has the highest number of records, followed by Lasalgaon and Pimpalgaon. The record count decreases steadily from Nagpur to Kolhapur, which has the lowest count. This suggests that data collection efforts were more concentrated in Nagpur and its surrounding markets, with decreasing coverage towards the markets with lower record counts.

Correlation Plot



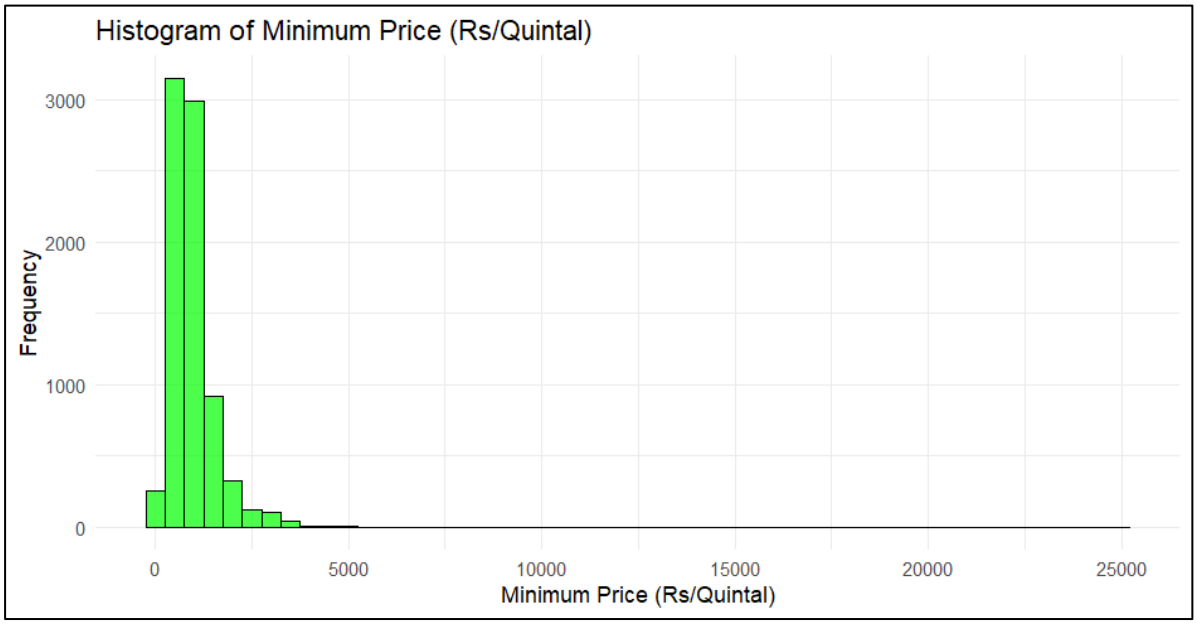
The heatmap shows the correlations between different price variables. Minimum and maximum prices have a strong positive correlation, and both are strongly positively correlated with modal price. Arrivals show a weak negative correlation with minimum price.



The pair plot shows relationships between prices. Minimum and maximum prices are strongly positively correlated, and both are strongly positively correlated with modal price. Arrivals show a weak negative correlation with minimum price.

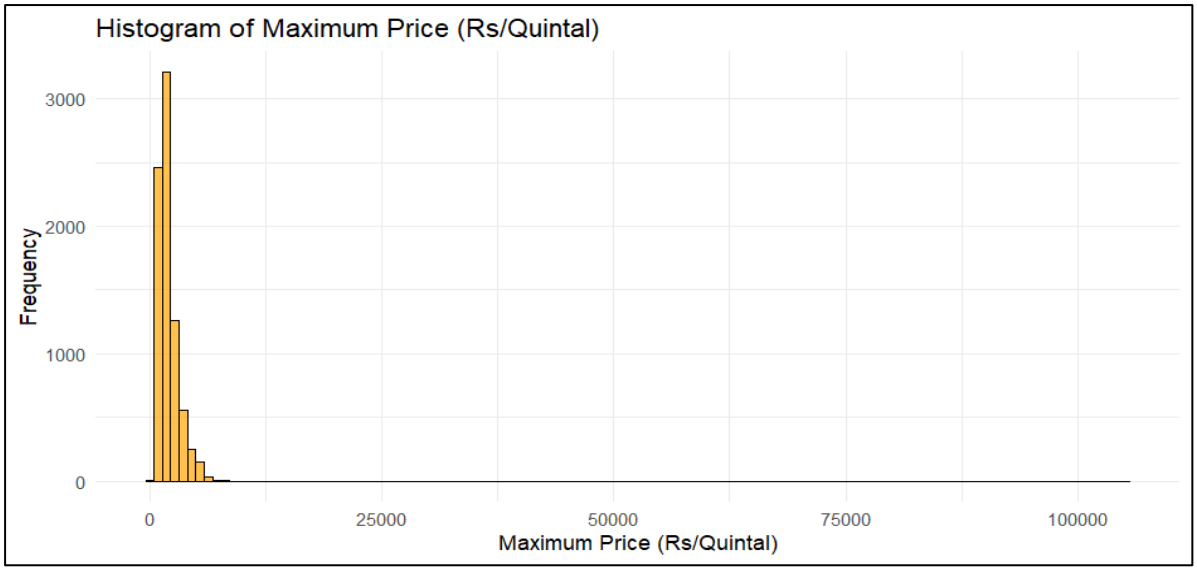
EDA On Pune City

Histogram for Minimum Price



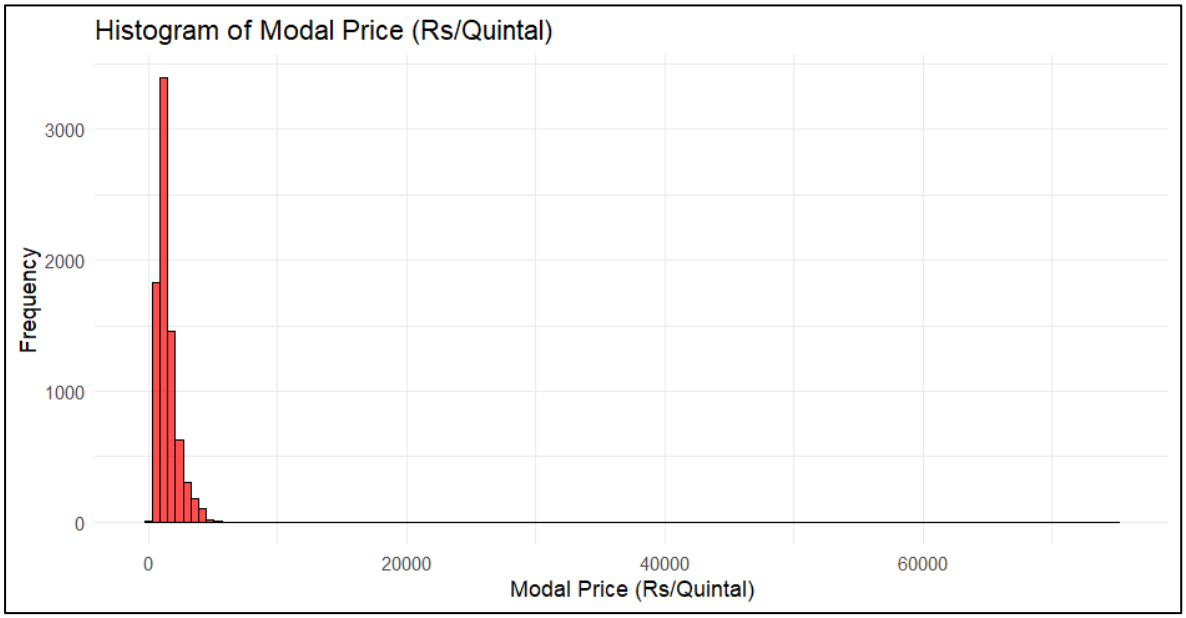
The histogram shows that the minimum prices are highly concentrated between 0 and 5000 Rs/Quintal, with a significant peak around 3000 Rs/Quintal. The distribution is right-skewed, indicating the presence of a few very high prices.

Histogram for Maximum Price



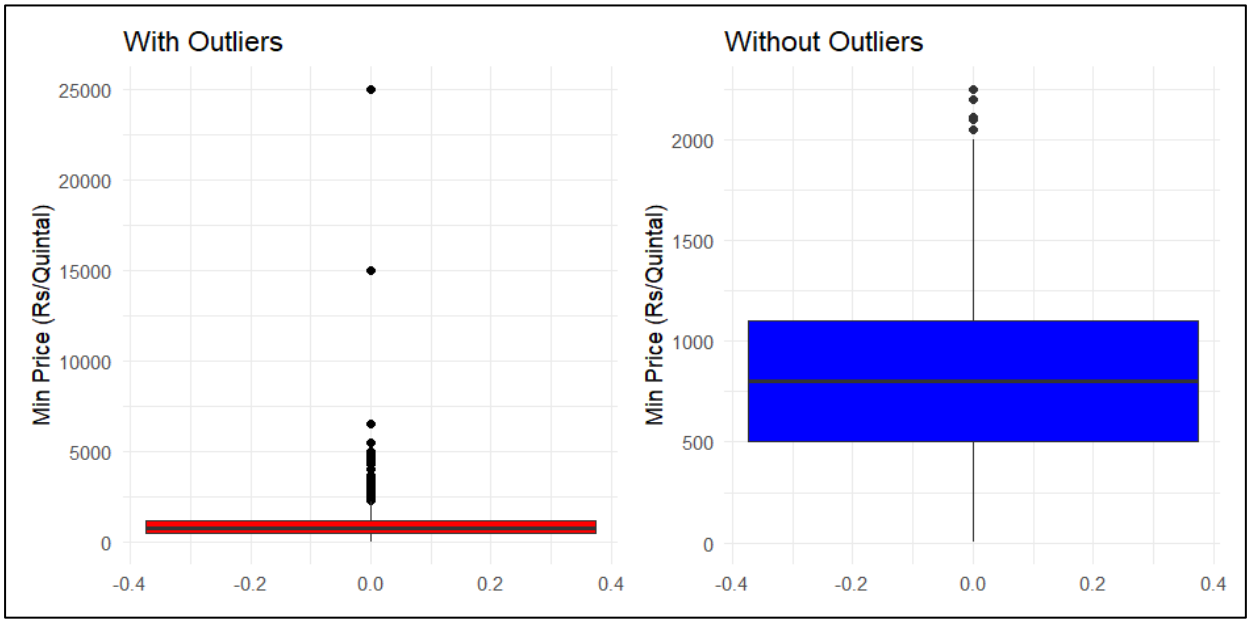
The histogram of maximum prices reveals that the majority of prices fall within the range of 0 to 10,000 Rs/Quintal, with a significant peak around 3,000 Rs/Quintal. The distribution is right-skewed, indicating the presence of a few very high maximum prices.

Histogram for Modal Price



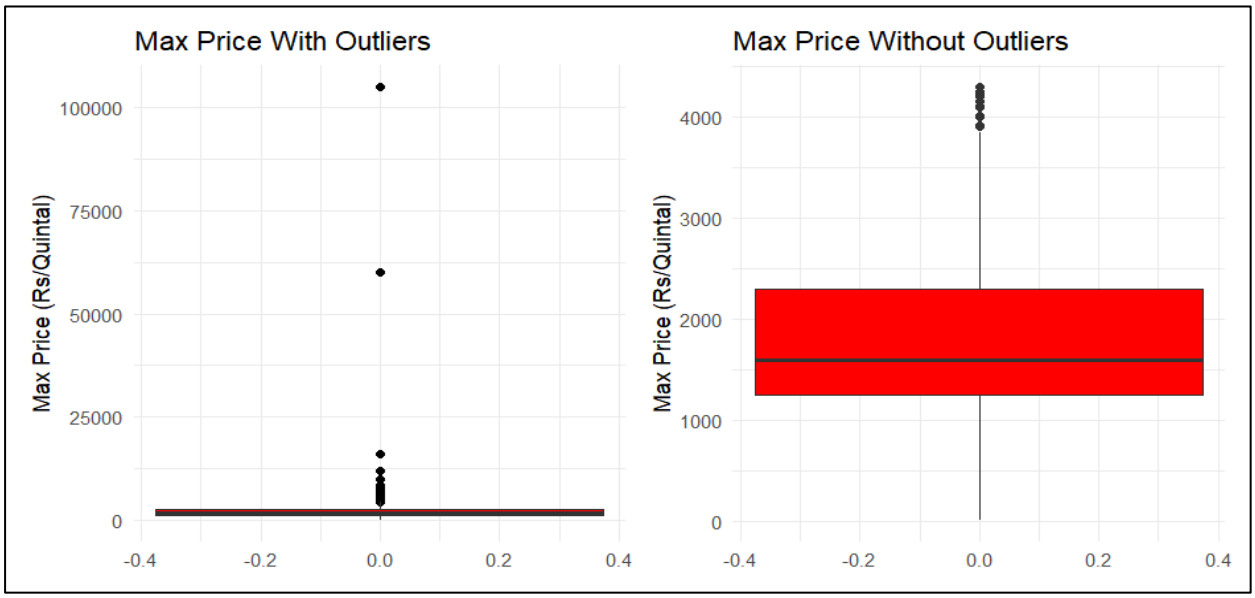
The histogram of modal prices shows that the majority of prices fall within the range of 0 to 10,000 Rs/Quintal, with a significant peak around 3,000 Rs/Quintal. The distribution is right-skewed, indicating the presence of a few very high modal prices.

Boxplot For Minimum Prices



Boxplots show the distribution of minimum prices (in Rs/Quintal).
With Outliers: The boxplot shows a few very high prices (outliers) pulling the right whisker up. The majority of prices are concentrated in the lower range, with a median around 1000 Rs/Quintal.
Without Outliers: Removing the outliers reveals a more symmetrical distribution. The median remains around 1000 Rs/Quintal, but the box is narrower, indicating less variability in the majority of prices.

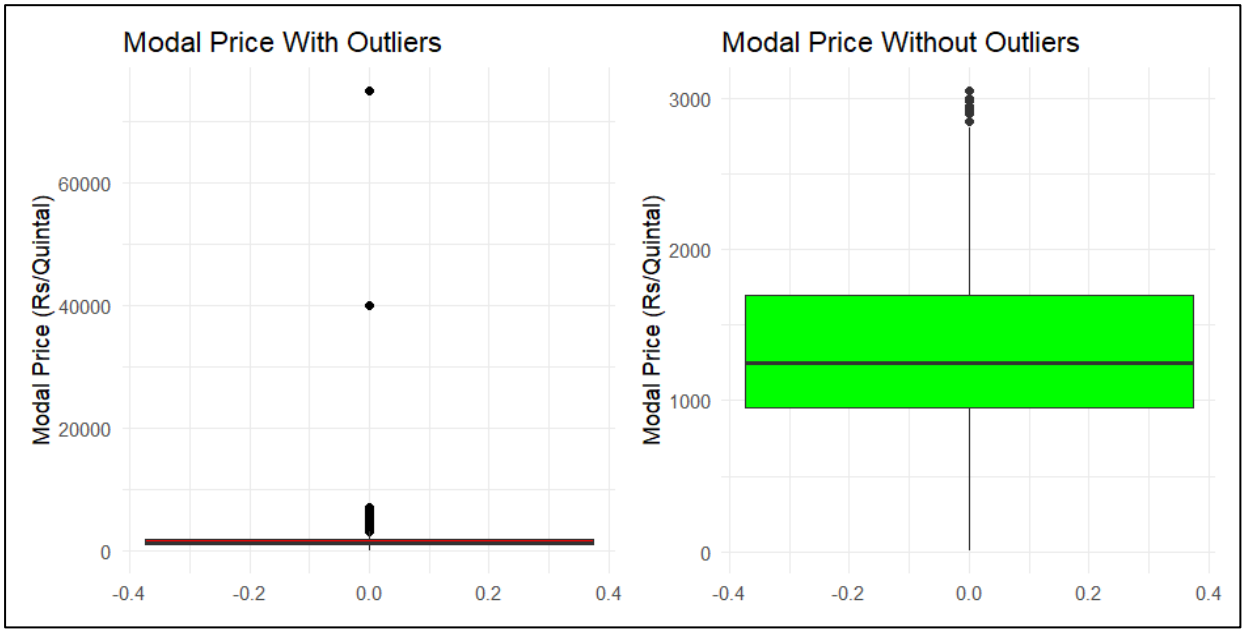
Boxplot For Maximum Prices



Boxplots show the distribution of maximum prices (in Rs/Quintal).

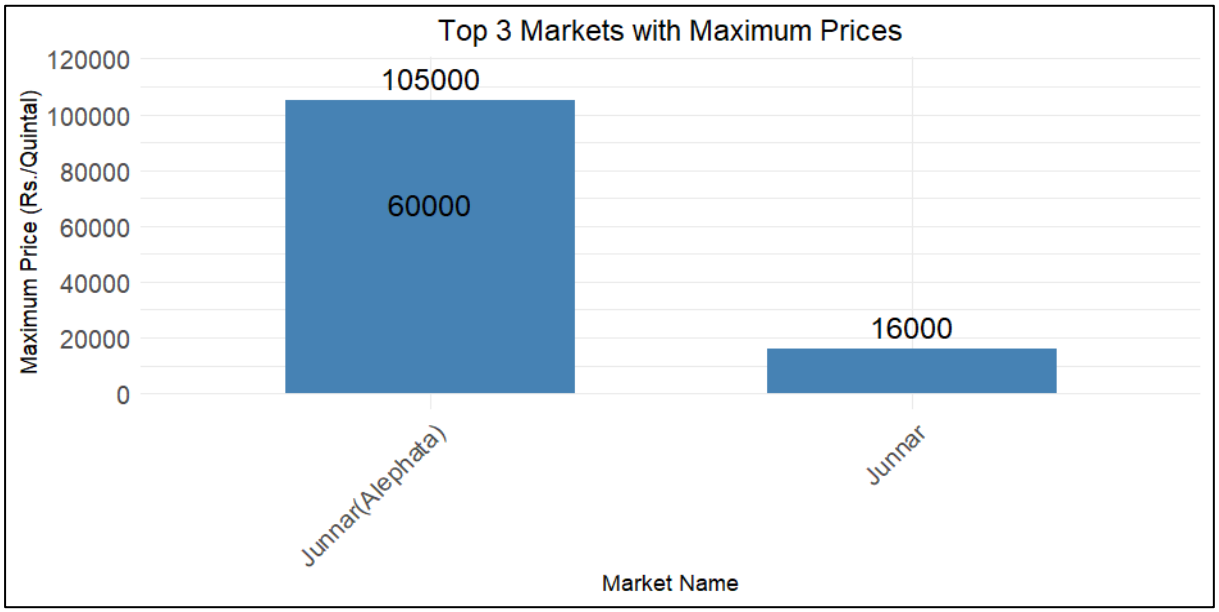
With Outliers: The boxplot shows a few very high prices (outliers) pulling the right whisker up. These outliers range from approximately **25,000 Rs/Quintal** to **100,000 Rs/Quintal**. The majority of prices are concentrated in the lower range, with a median around 2000 Rs/Quintal.
Without Outliers: Removing the outliers reveals a more symmetrical distribution. The median remains around 2000 Rs/Quintal, but the box is narrower, indicating less variability in the majority of prices.

Boxplot For Modal Prices



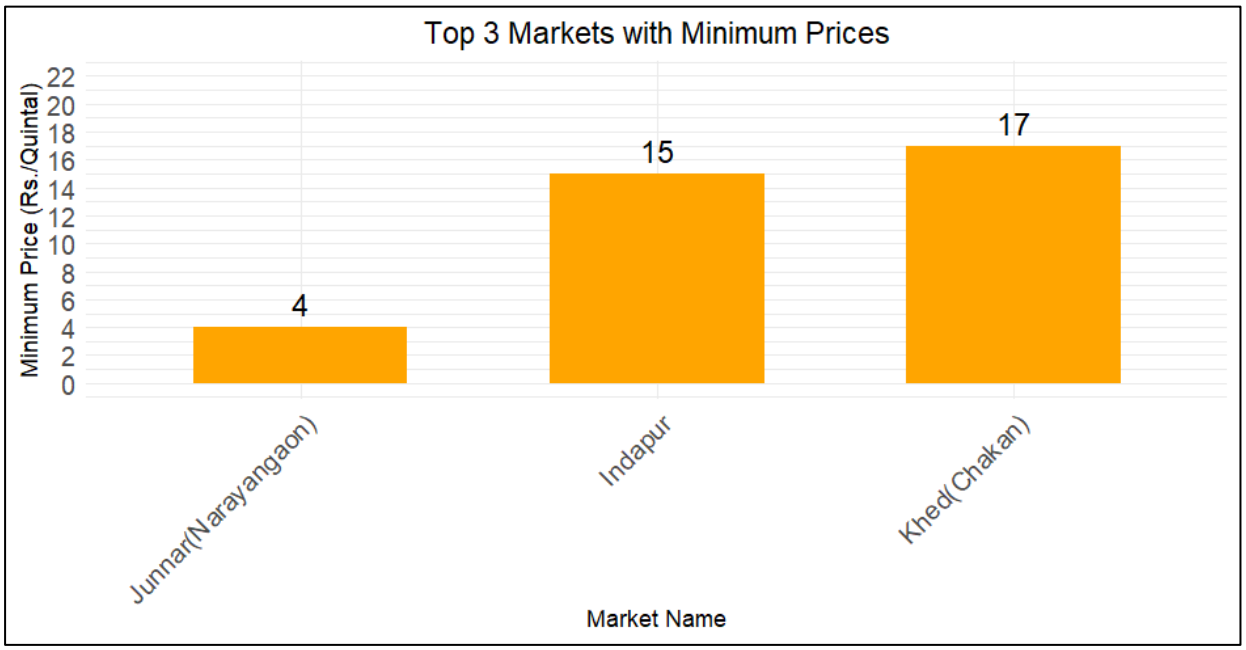
Boxplots show the distribution of modal prices (in Rs/Quintal).
With Outliers: The boxplot shows a few very high prices (outliers) pulling the right whisker up. These outliers range from approximately 40,000 Rs/Quintal to 70,000 Rs/Quintal. The majority of prices are concentrated in the lower range, with a median around 1000 Rs/Quintal.
Without Outliers: Removing the outliers reveals a more symmetrical distribution. The median remains around 1000 Rs/Quintal, but the box is narrower, indicating less variability in the majority of prices.

Top 3 Markets with Maximum Prices



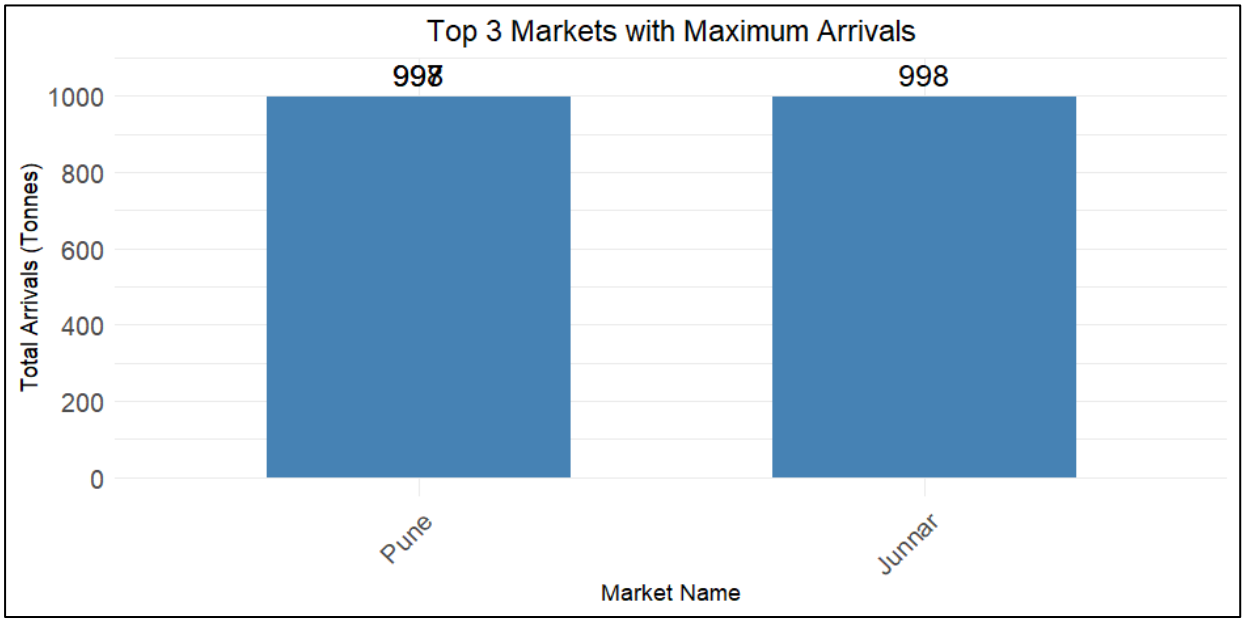
The bar chart shows the top 3 markets with the highest maximum prices. Junnar (Alephata) has the highest maximum price at 105,000 Rs./Quintal, followed by Junnar at 60,000 Rs./Quintal, and another market at 16,000 Rs./Quintal.

Top 3 Markets with Minimum Prices



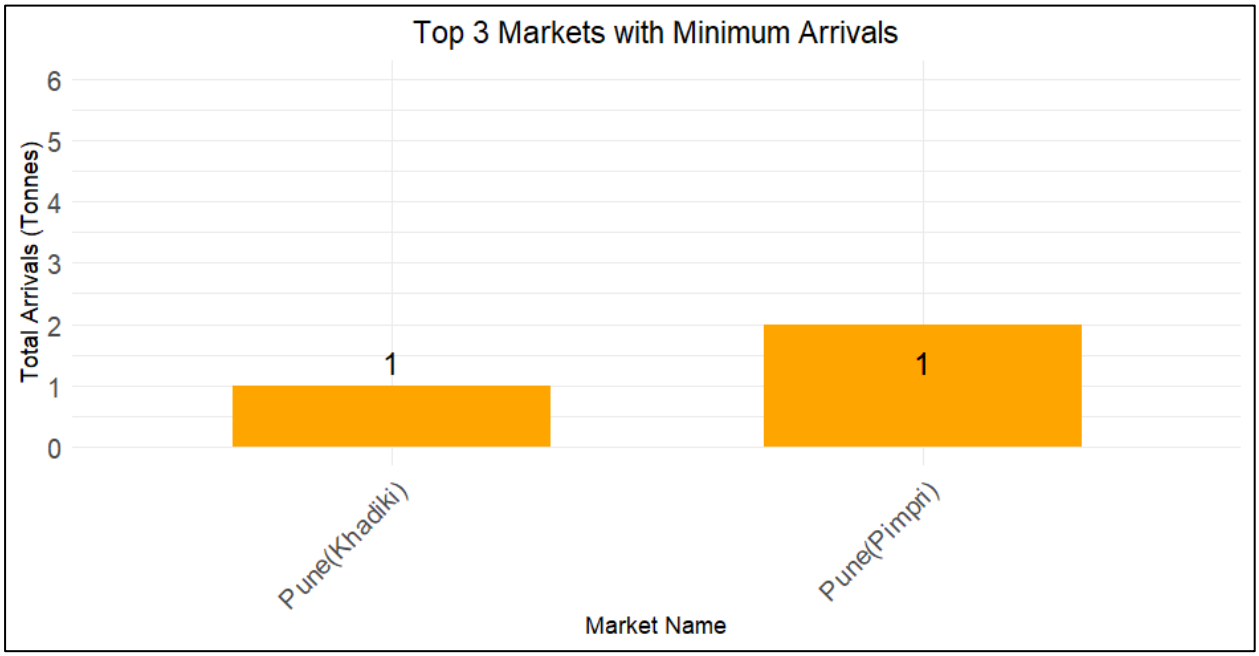
The bar chart shows the top 3 markets with the lowest minimum prices. Junnar (Narayangaon) has the lowest minimum price at 4 Rs./Quintal, followed by Indapur at 15 Rs./Quintal, and Khed (Chakan) at 17 Rs./Quintal.

Top 3 Markets with Maximum Arrivals



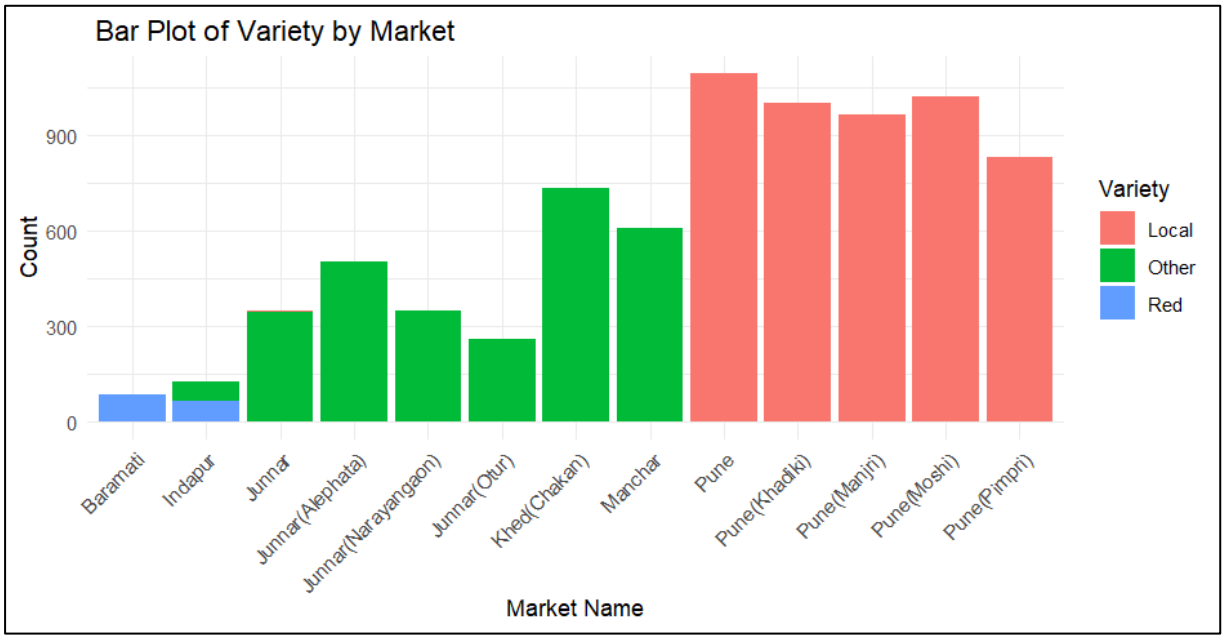
The bar chart shows the top 3 markets with the highest total arrivals. Pune and Junnar both have the highest total arrivals at 998 tonnes. Pune also have 997 tonnes total arrivals .

Top 3 Markets with Minimum Arrivals



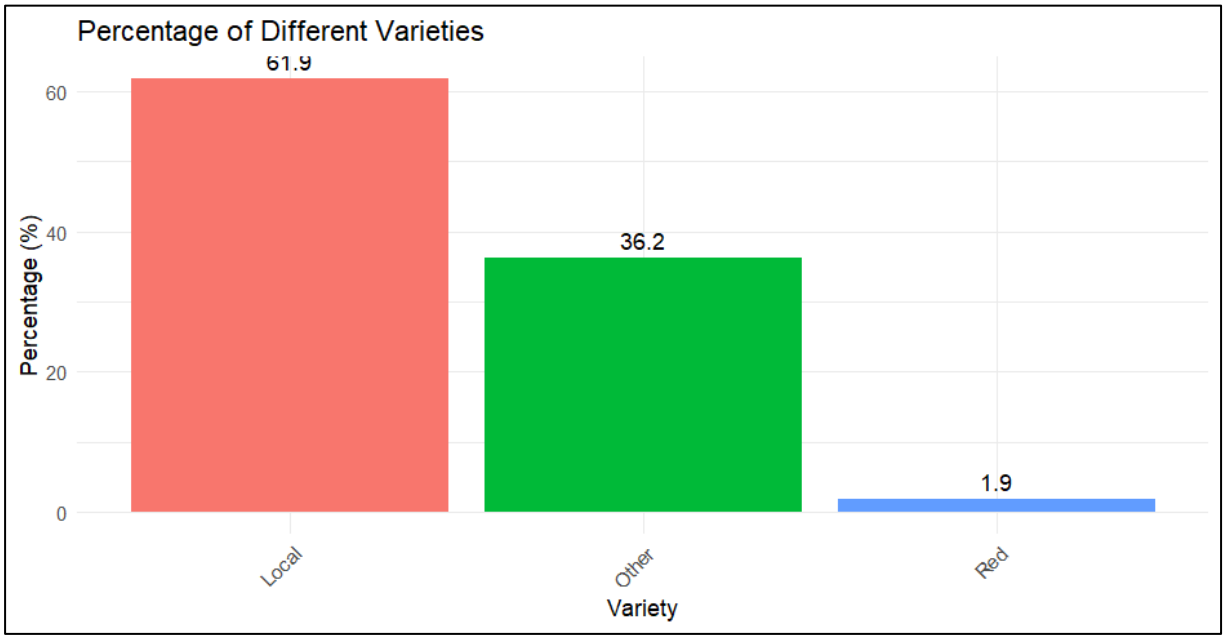
The bar chart shows the top 3 markets with the lowest total arrivals. Pune (Khadiki) and Pune (Pimpri) both have the lowest total arrivals at 1 tonne.

Bar Plot of Variety by Market



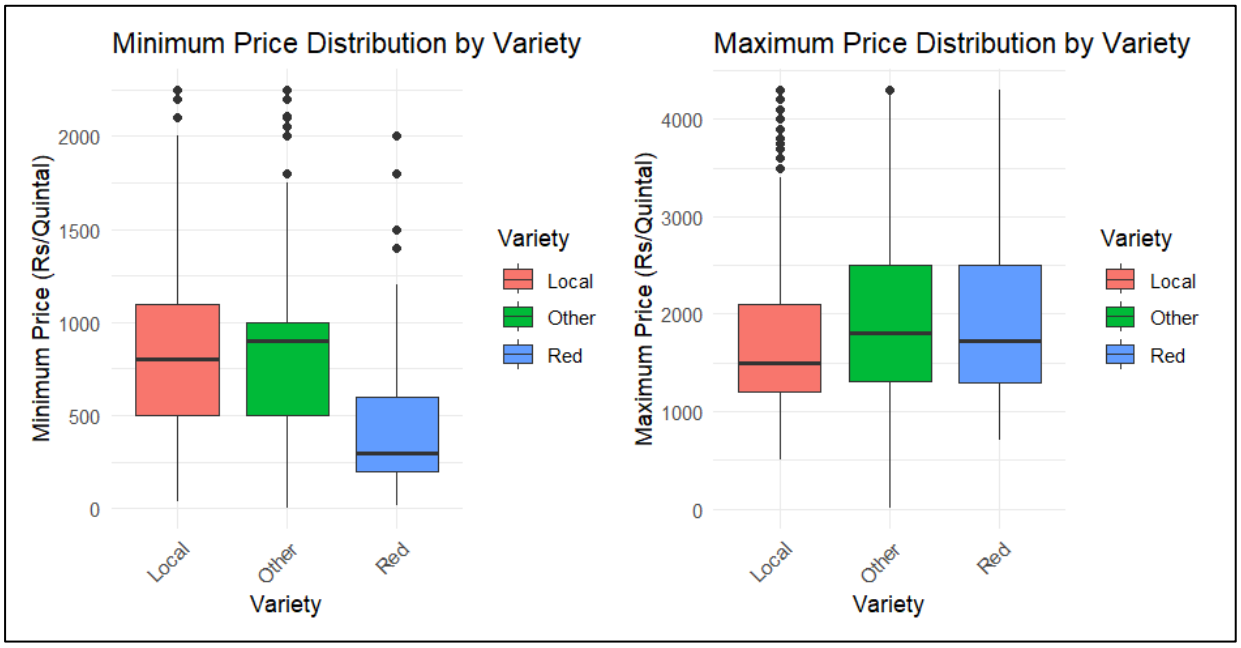
Pune markets dominate with the highest counts of "Local" varieties, while Junnar stands out with a significant proportion of "Other" varieties. Baramati and Indapur show a noticeable presence of "Red" varieties compared to other markets.

Bar Plot Calculate counts and percentages



The bar chart shows the percentage distribution of different varieties. "Local" varieties are the most prevalent, accounting for 61.9% of the total. "Other" varieties comprise 36.2%, while "Red" varieties constitute a minor 1.9%.

Box Plot of Max and Min Price by Variety



The box plots reveal that "Other" varieties exhibit a wider range of maximum prices, while "Local" varieties show greater variability in minimum prices. Notably, "Other" varieties tend to have higher median maximum prices, whereas "Local" varieties generally command higher median minimum prices.

Comparison of Prices & Arrivals of Pre Covid and Post Covid Period

We divide the whole dataset into two parts,pre covid-19 (1-1-2013 to 23-3-2020) and remaining as post covid-19 (15-6-2021 to 31-12-2024) . we compare the prices after covid and before covid. And during Covid-19 Period. Aslo we use t test for comparision

In Maharashtra, the lockdown periods during the COVID-19 pandemic were as follows:
First Lockdown: March 23, 2020 – June 1, 2020
Second Lockdown: April 15, 2021 – June 15, 2021

Welch's t-test: Theory and Application
Welch's t-test is a statistical method used to compare the means of two independent groups when the assumption of equal variances (homoscedasticity) is violated. Unlike the traditional Student's t-test, Welch's t-test is more robust as it does not require the variances of the two groups to be equal. This makes it particularly useful in real-world scenarios where group variances and sample sizes often differ.
Key Features:

1. **Adjustment for Variance**: Welch's t-test accounts for unequal variances between groups by modifying the test statistic and degrees of freedom.
2. **Flexible Assumptions**: It does not assume equal sample sizes or variances, making it versatile for diverse datasets.
3. **Hypotheses**:
   - Null Hypothesis (H0): There is no difference in the population means ($\mu1=\mu2$).
   - Alternative Hypothesis (Ha): There is a difference in the population means ($\mu1\neq\mu2$, $\mu1>\mu2$, or $\mu1<\mu2$).

Formula:
The test statistic for Welch's t-test is calculated as:
Where:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- $\bar{x}_1, \bar{x}_2$ are the sample means.
- s_1^2, s_2^2 are the sample variances.
- n_1 ,n_2 are the sample sizes.

The degrees of freedom (df) are estimated using the Welch-Satterthwaite equation:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$$

Interpretation:

- **t-value**: Indicates the magnitude and direction of the difference between the two sample means.
- **p-value**: Measures the probability of observing the data under the null hypothesis. A smaller p-value (e.g., < 0.05) indicates a significant difference in means.
- **Confidence Interval**: Provides a range of values within which the true mean difference is likely to fall.

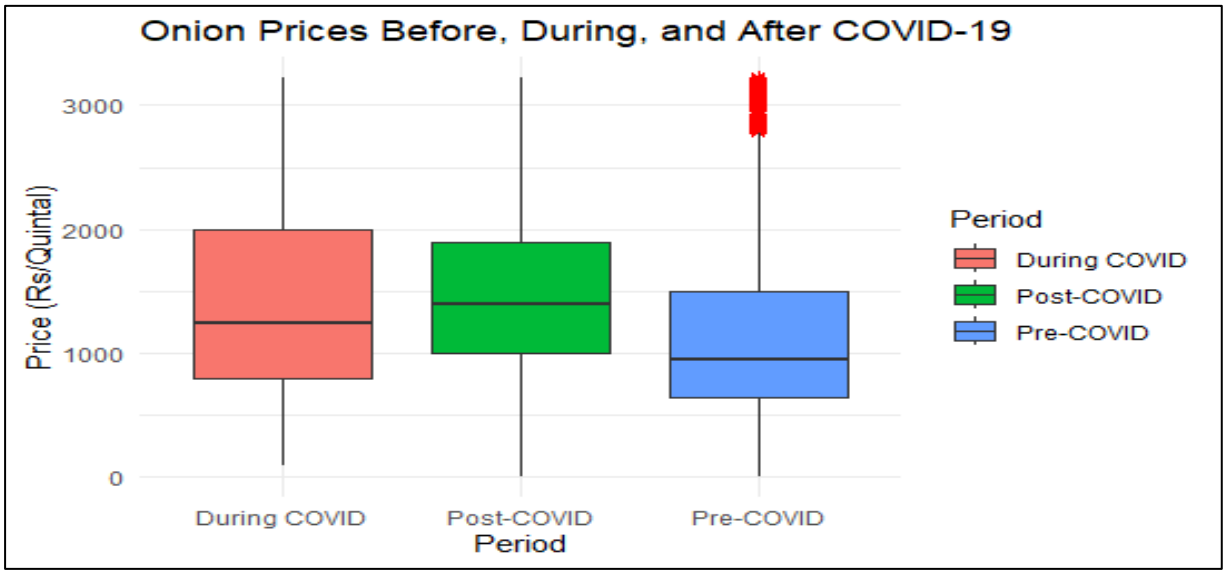
Advantages:

- Robust to unequal variances and sample sizes.
- Provides accurate results in scenarios where assumptions of the Student's t-test are violated.

Applications:

Welch's t-test is widely used in fields like medicine, social sciences, and economics to analyze differences between two groups, such as treatment vs. control groups or pre- and post-intervention groups.

Boxplots of Onion Price Before, During, and After Covid-19



This boxplot compares onion prices across Pre-COVID, During COVID, and Post-COVID periods:

Pre-COVID: Prices were lower and. more stable, with a few outliers.

During COVID: Prices surged with higher variability due to disruptions like lockdowns and labor shortages.

Post-COVID: Prices remained elevated but showed reduced variability, reflecting market recovery and stabilization.

Onion prices increased significantly during COVID and remained high post-pandemic, driven by supply chain disruptions and inflation, though variability decreased as markets stabilized.

T- Test For Comparison :

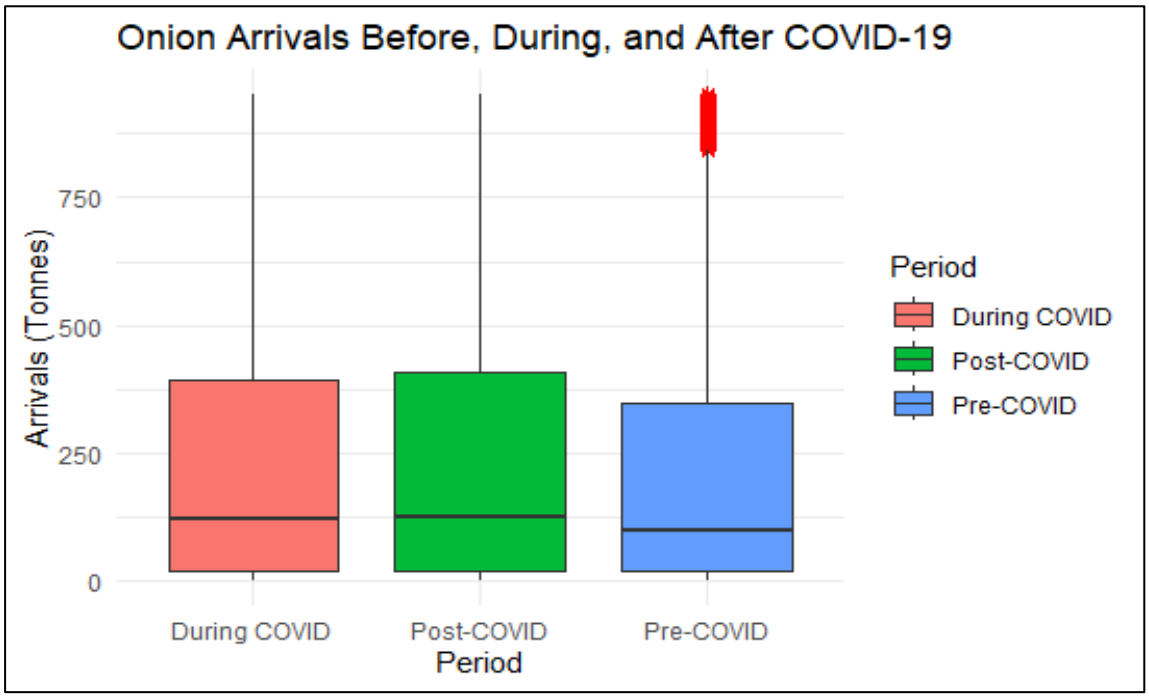
T-Test: During COVID vs Pre-COVID

The mean onion price during COVID (₹1454/quintal) was significantly higher than pre-COVID (₹1152/quintal). p-value is $< 2.2e-16$, indicating a statistically significant difference. COVID had a significant impact, leading to a price increase.

T-Test: During COVID vs Post-COVID

The mean price **during COVID (₹1454/quintal)** was slightly lower than **post-COVID (₹1506/quintal). p-value is 1.42e-12**, indicating a statistically significant difference. Prices remained high **post-COVID** but with a small upward shift compared to during COVID.

For Arrivals



The boxplot shows onion arrivals across three periods: **Pre-COVID**, **During COVID**, and **Post-COVID**. Median arrivals are around 250 tonnes for all periods, with slightly higher values during COVID. Variability (IQR) is highest Pre-COVID (approx. 100–400 tonnes), compared to narrower ranges During COVID (100–350 tonnes) and Post-COVID (150–350 tonnes). Outliers above 750 tonnes are observed only in the Pre-COVID period, indicating greater variability. Arrivals were more stable during and after COVID.

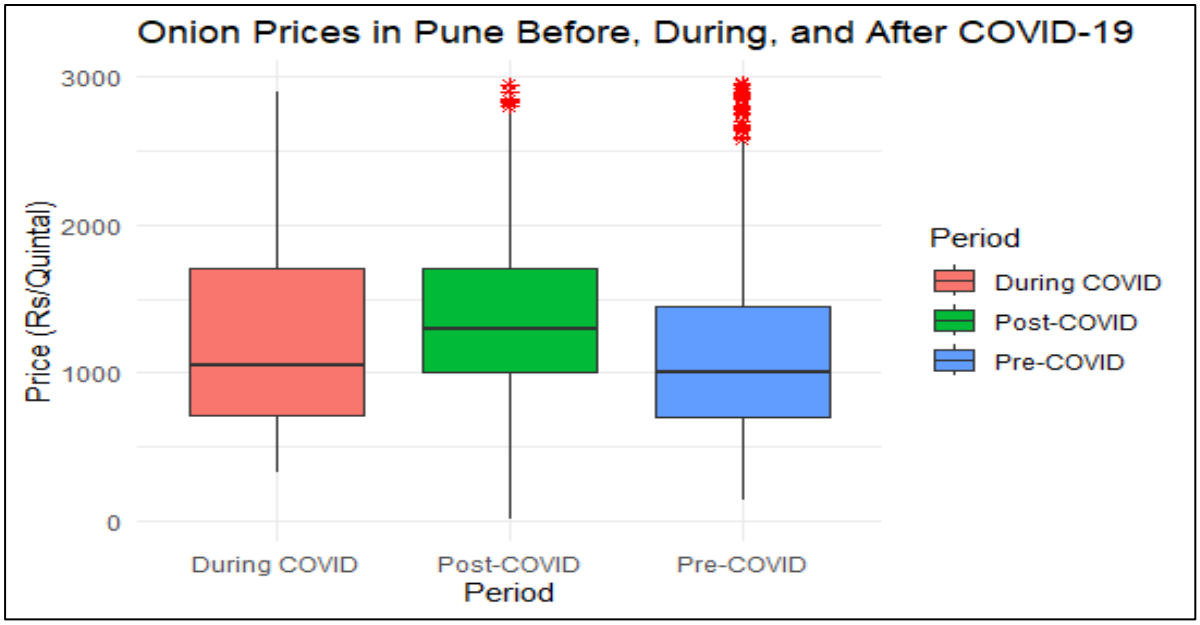
T- Test For Comparison :-

Two Welch t-tests were conducted:

- **During COVID vs. Pre-COVID:** The mean arrivals during COVID (233 tonnes) were significantly higher than Pre-COVID (217 tonnes), with a p-value of 6.15×10−96.15 \times 10^{-9}6.15×10−9, indicating a statistically significant difference.
- **During COVID vs. Post-COVID:** The mean arrivals during COVID (233 tonnes) were significantly lower than Post-COVID (241 tonnes), with a p-value of 0.003, showing a statistically significant difference.

Comparison for Pune District

For Prices



Pre-COVID: Prices have the widest spread, with outliers above ₹3000/quintal, indicating high price volatility.

During COVID: Prices were more stable with a lower median compared to other periods.

Post-COVID: Prices increased slightly compared to the COVID period, with a narrower range but fewer outliers.

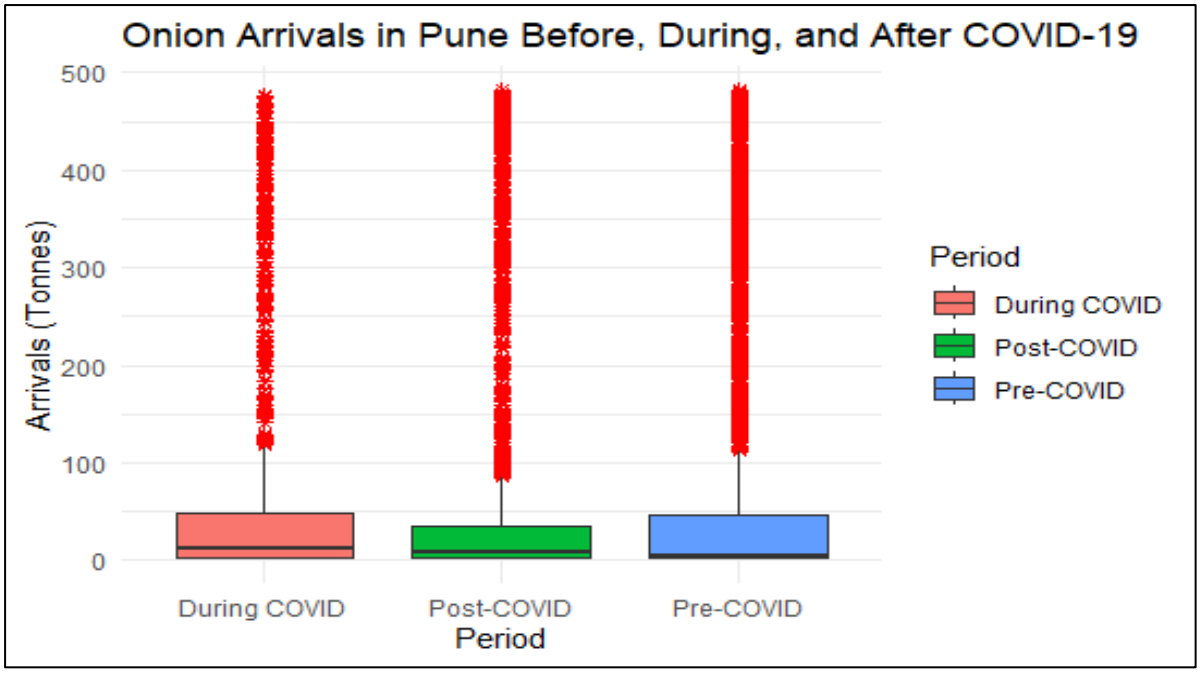
T Test :-

During COVID vs Pre-COVID: The mean price during COVID (₹1264.79/quintal) was significantly higher than pre-COVID (₹1134.55/quintal) with a p-value of 1.249e-11.

During COVID vs Post-COVID: The price during COVID (₹1264.79/quintal) was significantly lower than post-COVID (₹1413.93/quintal) with a p-value of 1.913e-15.

Onion prices increased significantly from **pre-COVID** to **during COVID**, then further rose **post-COVID.**

For Arrivals



This boxplot shows **onion arrivals** in Pune were similar **before, during, and after COVID-19**. While typical **arrival amounts** (the boxes) stayed roughly the same, there were many days with unusually high arrivals (the dots/outliers) in all three periods, suggesting these spikes were not related to the pandemic. Onion arrivals in Pune were generally consistent (around 50 tonnes, ranging roughly from 50-480 tonnes) before, during, and after COVID-19. However, there were frequent, unusually **high arrival** days in all periods, unrelated to the pandemic.

T Test:-

During COVID vs Pre-COVID: The mean arrivals during COVID (66.58 tonnes) are slightly higher than pre-COVID (64.30 tonnes), but the p-value (0.5759) indicates no statistically significant difference.

During COVID vs Post-COVID: The mean arrivals during COVID (66.58 tonnes) are significantly higher than post-COVID (39.76 tonnes), with a p-value of 1.446e-12, indicating a statistically significant difference.

There was no significant difference in arrivals between during COVID and pre-COVID, but there was a significant increase in arrivals during COVID compared to post-COVID.

Time Series Analysis

SARIMA

SARIMA, which stands for Seasonal Autoregressive Integrated Moving Average, is a versatile and widely used time series forecasting model. It's an extension of the non-seasonal ARIMA model, designed to handle data with seasonal patterns.

SARIMA captures both short-term and long-term dependencies within the data, making it a robust tool for forecasting. It combines the concepts of autoregressive (AR), integrated (I), and moving average (MA) models with seasonal components.

The SARIMA Notation:

SARIMA(p, d, q)(P, D, Q, s),

- AR(p): Autoregressive component of order p,
- MA(q): Moving average component of order q,
- I(d): Integrated component of order d,
- Seasonal AR(P): Seasonal autoregressive component of order P,
- MA(Q): Seasonal moving average component of order Q,
- Seasonal I(D): Seasonal integrated component of order D,
- s: Seasonal period

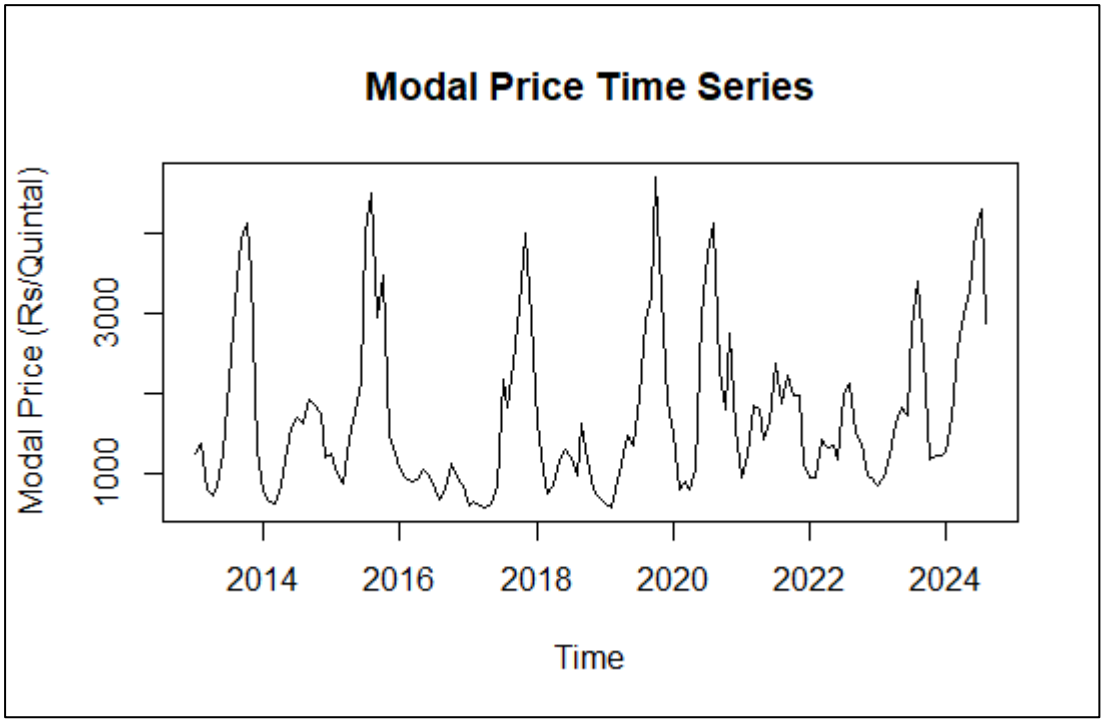
The mathematical representation of SARIMA is as follows:

Where,

$$(1 - \phi)B(1 - \Delta B)(1 - B)(1 - \Delta P) = \mu(1 + B)(1 + \Delta B)(Pf)$$

- yt = the observed time series at time t
- $\Phi 1$ = is the non-seasonal autoregressive coefficient,
- $\Theta 1$ = is the non-seasonal moving average coefficient,
- $\theta 1$ = is the seasonal moving average coefficient,
- ϵt = is the white noise error term at time t.

Autoregressive (AR) Component: The autoregressive non-seasonal component represented by captures the relationship between the current observation and a certain number of lagged observations (previous values in the time series). The B term represents the backshift operator is commonly used in time series analysis. It represents the lag operator, which shifts the time series backward by a certain number of time period. The order of the autoregressive component, denoted by (p), determines the number of past values considered in the model.

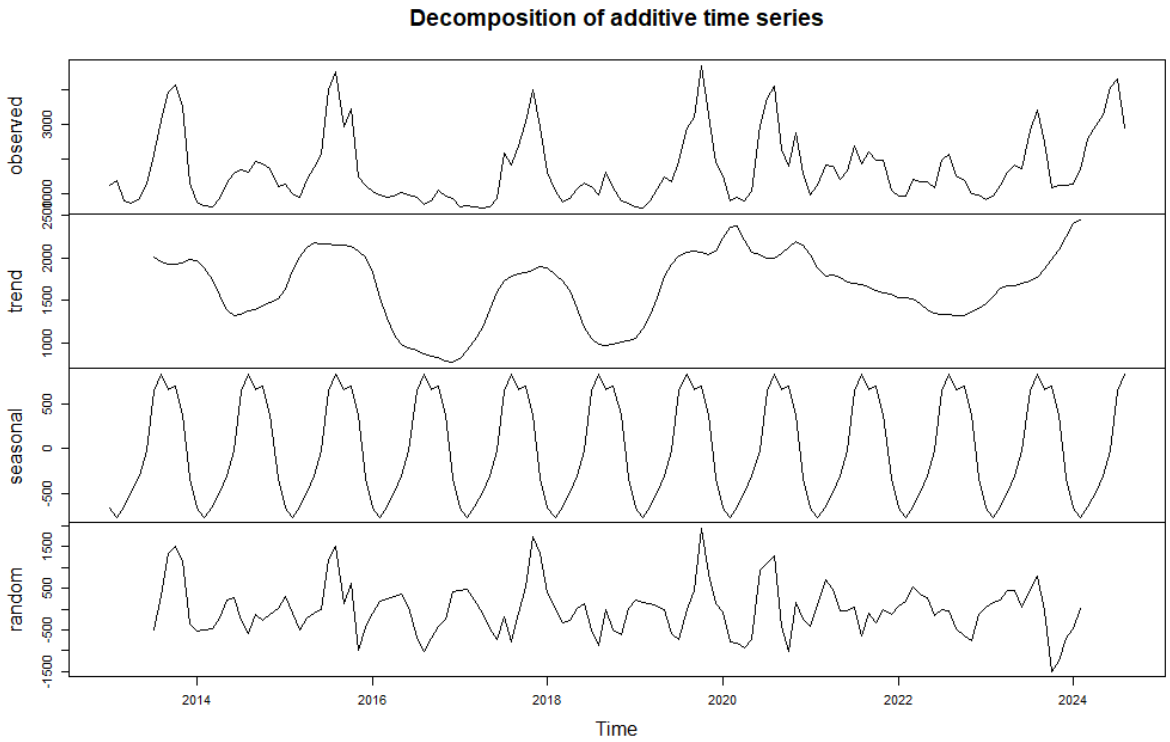


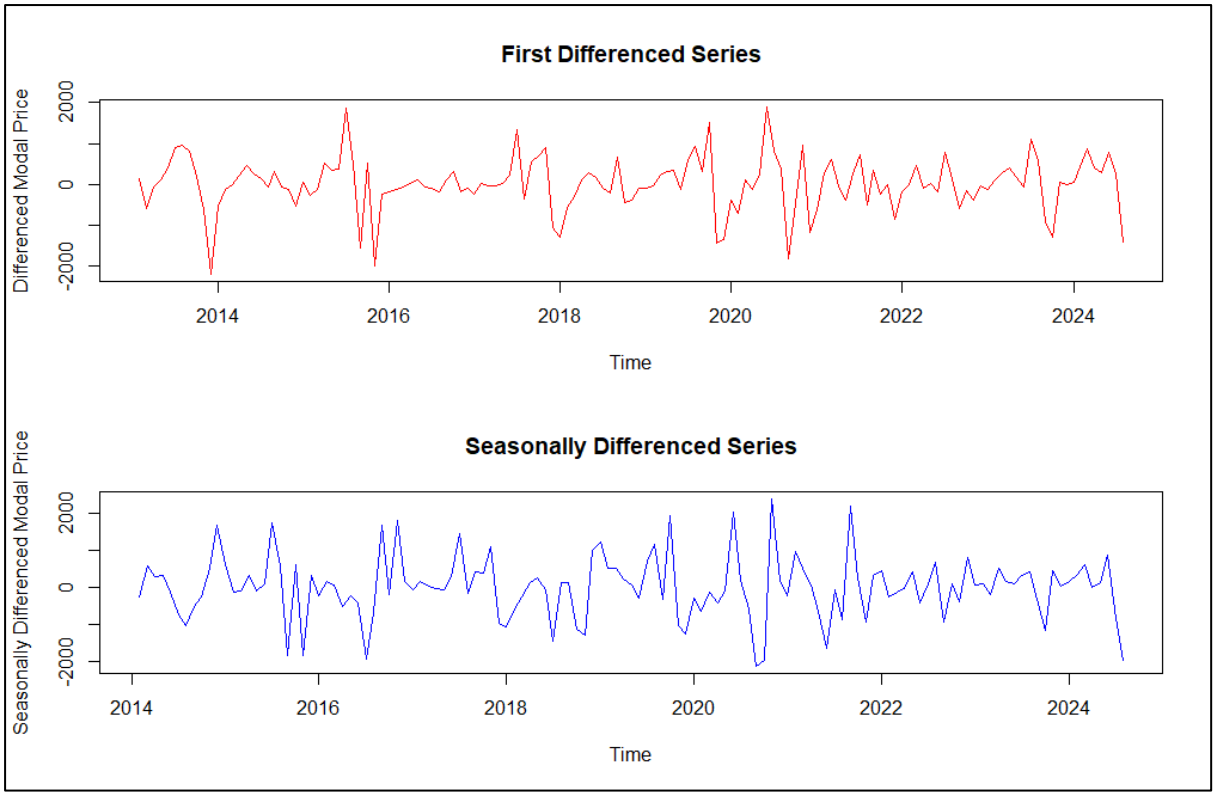
The plot reveals a cyclical pattern in onion prices, with recurring peaks and troughs over roughly two-year intervals. The prices exhibit significant volatility, fluctuating rapidly between highs and lows. The series appears to be non-stationary due to variations in the mean and variance over time.

Further, this data set is decomposed for a better understanding of the time series. In this study, classical decomposition by an additive method with a seasonal period of 12 is used**.**

The decomposition of the modal price time series for onions in the Pune-Junnar region (2013–2024) reveals key components: observed, trend, seasonal, and random. The trend shows a general increase in prices over time, with a dip around 2018–2020 likely due to external factors. The seasonal component highlights regular fluctuations driven by harvest cycles and demand patterns. The random component reflects irregular variations, such as market shocks or supply disruptions. This analysis confirms the influence of seasonality and trends on prices, making forecasting methods like SARIMA or Holt-Winters suitable for predicting future price movements**.**

In order to apply according to Box–Jenkins method, the given time series must be stationary. To test the stationarity of the Modal price of onion, the augmented Dickey–Fuller test is conducted. The hypothesis of the augmented Dickey–Fuller test describes as follows: • Null hypothesis: Time series (Modal price) contains a unit root or non-stationary and expressed as follows: $H0 : \Phi = 1$ • Alternate hypothesis: Time series (Modal price) contains stationary and expressed as follows: $HA : \Phi < 1$


Decomposition of additive time series
observed
trend
seasonal
random
Time
2014
2016
2018
2020
2022
2024



The p value for augmented Dickey-Fuller test is 0.01 which is less than 0.05, Reject Null Hypothesis (H0), i.e. time series does not have a unit root,meaning it is stationary. It does not have a time-dependent structure.

Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test:

Null Hypothesis (H0): Series is trend stationary or series has no unit root.

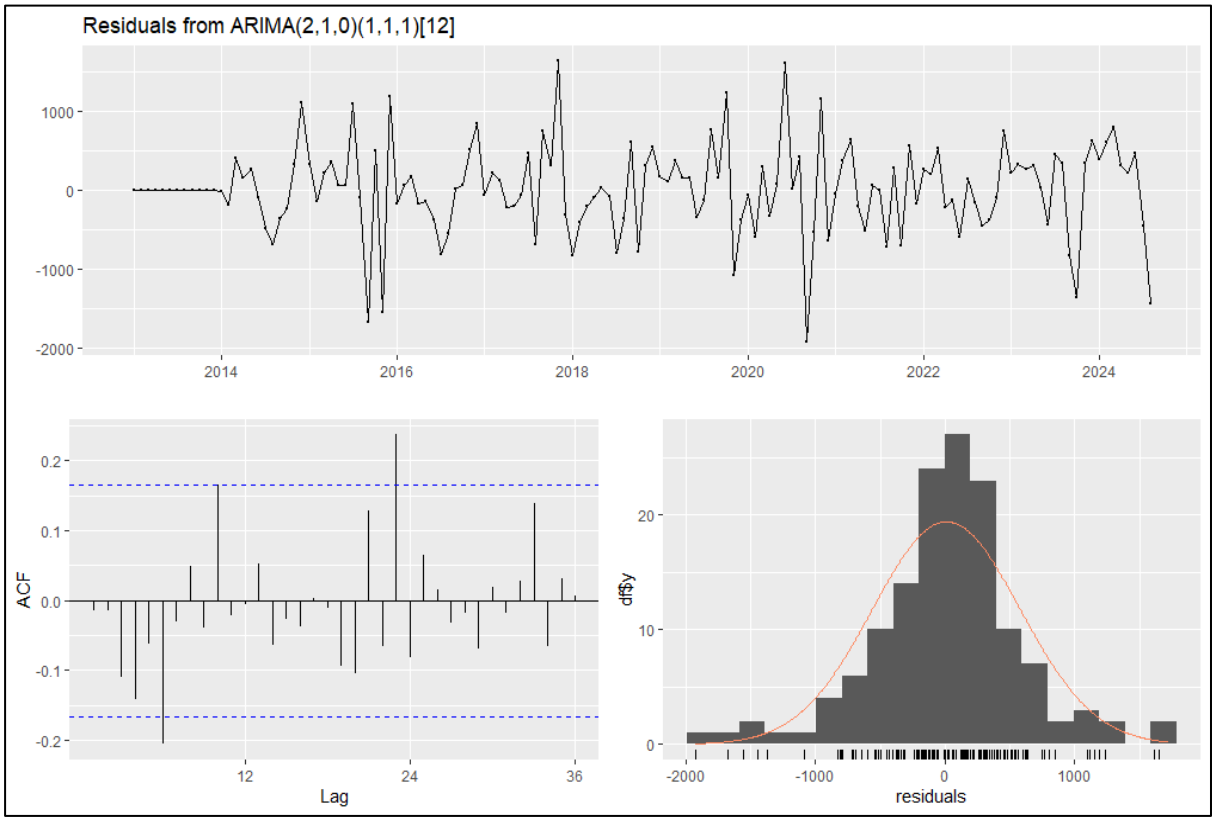
Alternate Hypothesis(HA): Series is non-stationary, or series has a unit root.

If the null hypothesis is failed to be rejected, this test may provide evidence that the series is trend stationary.

Conditions to Fail to Reject Null Hypothesis(H0)

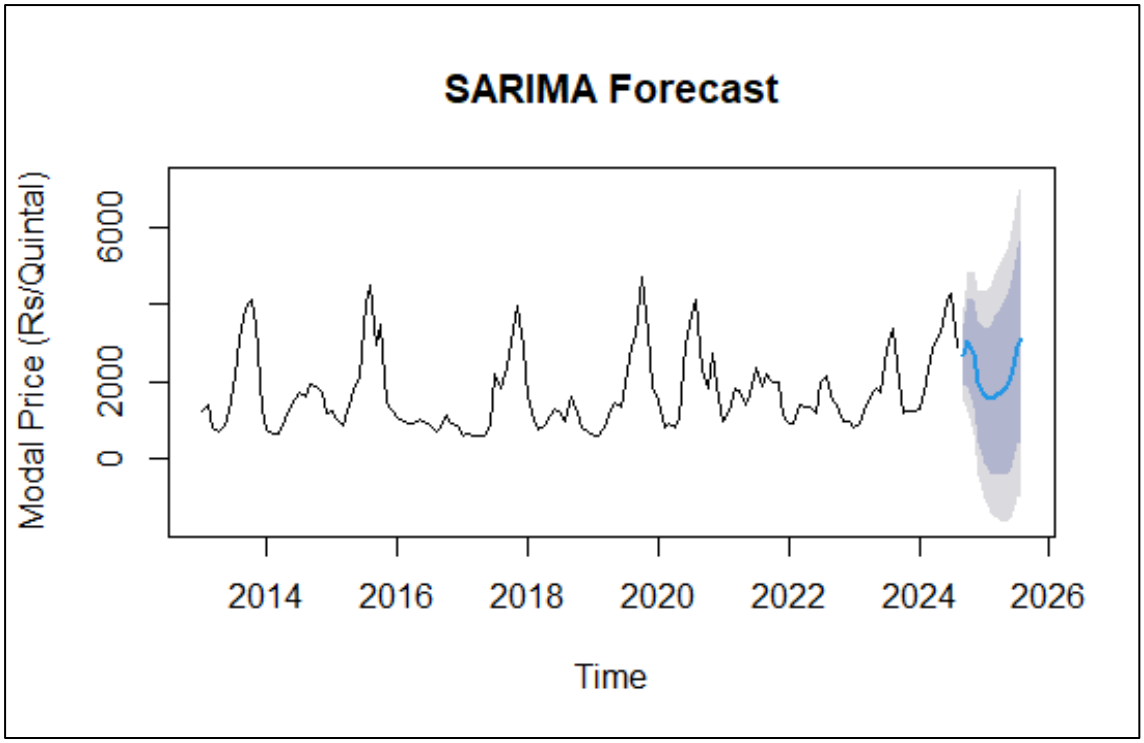
If the Test Statistic Critical Value and p-value is less than 0.05 then Fail to Reject Null Hypothesis(H0), i.e., time series does not have a unit root, meaning it is trend stationary. The KPSS value for our data comes out to be 0.01 which is less than 0.05 thus time series is stationary.

To check the autocorrelation between residuals we used Box-Ljung Test. P value for this test is 0.01769, which is less than chosen significant value, so there is autocorrelation in the series



This plot shows the residual diagnostics for the ARIMA(2,1,0)(1,1,1)[12] model applied to onion prices in the Pune-Junnar region:

1. Residual Plot: Residuals fluctuate around zero with no clear pattern, indicating the model captures the main structure of the data, though occasional outliers exist.
2. ACF: Most lags fall within the confidence bounds, suggesting residuals are nearly white noise with minimal unexplained dependencies.
3. Histogram: Residuals are roughly normally distributed, centered around zero, with slight deviations in the tails.

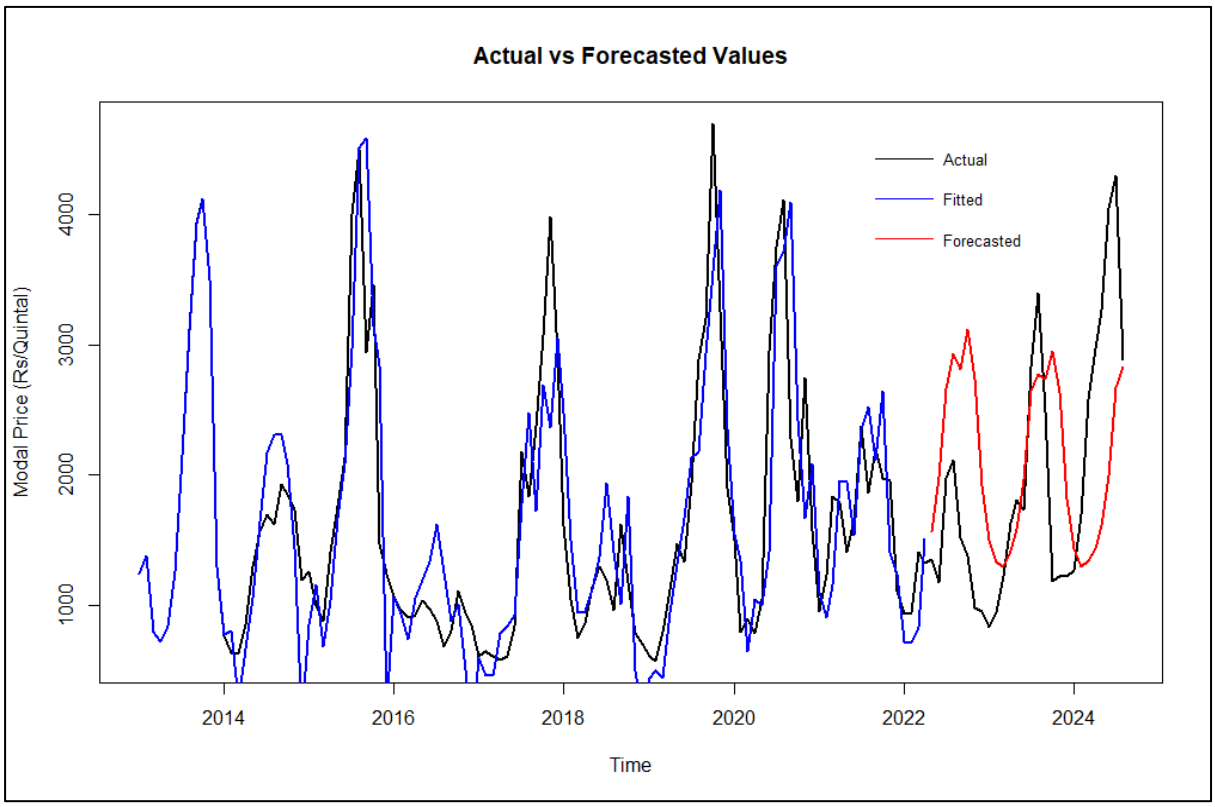


The graph shows the SARIMA model's forecast for onion prices (Rs/Quintal) in the Pune-Junnar region. The black line represents historical prices, highlighting seasonal fluctuations and trends, while the blue line shows predictions for 2025–2026.

The shaded regions indicate confidence intervals, with darker and lighter areas representing 80% and 95% confidence levels, respectively, reflecting increasing uncertainty over time. This forecast captures seasonal patterns and provides insights for planning and decision-making in the onion market while accounting for variability in price predictions.

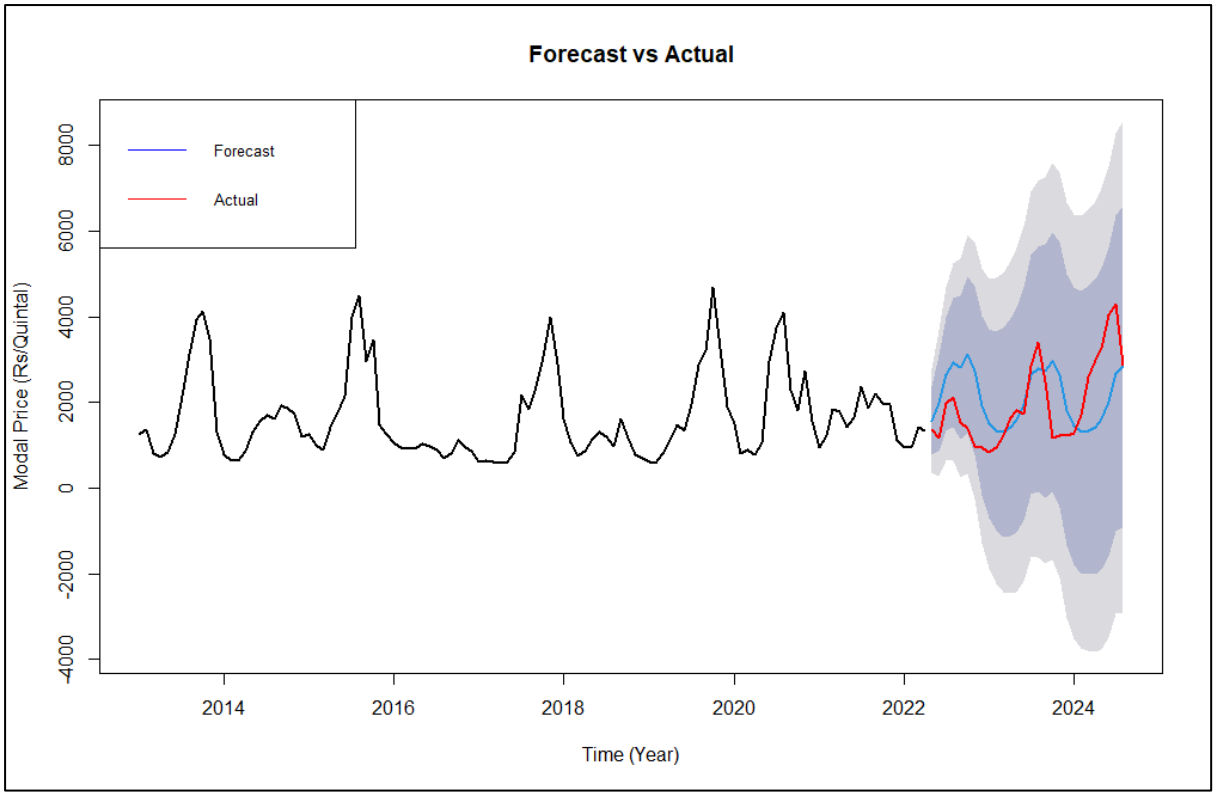
Data set division: The data set dive into training data and testing data .80% data for training and 20 percent for testing.

```
> # Fit SARIMA model to the training data
> sarima_model = Arima(train_data, order = c(2, 1, 0), seasonal = c(1, 1, 1))
> sarima_model
Series: train_data
ARIMA(2,1,0)(1,1,1)[12]

Coefficients:
          ar1      ar2     sar1     sma1
      -0.0152  -0.0746  -0.1736  -1.0000
s.e.   0.1020   0.1006   0.1072   0.1705

sigma^2 = 341294:  log likelihood = -784.34
AIC=1578.69   AICc=1579.33   BIC=1591.66
```



The plot compares actual, fitted, and forecasted modal prices using a SARIMA model. The fitted values closely align with actual prices, demonstrating the model's ability to capture trends and seasonality. The forecast continues the observed seasonal patterns, providing valuable insights for predicting future prices

It can be seen that there is no significantly much difference in fitted and actual Modal Price indicating our fitted SARIMA model is good.



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This plot compares the actual modal prices (red line) with the forecasted values (blue line) using a SARIMA model. The forecast captures the seasonal trends and variability seen in the historical data, with shaded confidence intervals indicating uncertainty. The close alignment between actual and forecasted values demonstrates the model's effectiveness in predicting future price movements while accounting for seasonal patterns. The model fits well, as evident from its ability to follow the trends in actual data, suggesting it is suitable for time series forecasting in this context.

Holt-Winters multiplicative model

The **Holt-Winters multiplicative model** is a popular method for forecasting time series data that exhibits both **trend** and **seasonality**. This method is based on **exponential smoothing** and assumes that the seasonal variations in the data change proportionally (multiplicatively) with the trend.

Theory of Holt-Winters Multiplicative Model

The multiplicative model captures three main components of a time series:

1. **Level** (Lt): The baseline value of the series after removing trend and seasonal effects.
2. **Trend** (Tt): The increasing or decreasing pattern in the data.
3. **Seasonality** (St): The repeating pattern in the data over a fixed period.

The multiplicative model assumes that the observed value (yt) is the product of these three components:

$$y_t = L_t \cdot T_t \cdot S_{t-m}$$

Here, m is the seasonal period (e.g., m=12 for monthly data with yearly seasonality).

Updating Equations

The Holt-Winters multiplicative model uses the following recursive equations to update the components at each time step:

1. **Level Component (LtL_tLt):**

$$L_t = \alpha \cdot \frac{y_t}{S_{t-m}} + (1-\alpha) \cdot (L_{t-1} \cdot T_{t-1})$$

- LtL_tLt: Smoothed level of the series at time ttt.
- α: Smoothing parameter for the level ($0 < \alpha \leq 1$).
- yt: Observed value at time t
- St−m: Seasonal factor for the same season in the previous period.

2. **Trend Component (Tt):**

$$T_t = \beta \cdot \frac{L_t}{L_{t-1}} + (1-\beta) \cdot T_{t-1}$$

- Tt: Smoothed trend at time t.
- β: Smoothing parameter for the trend ($0 < \beta \leq 1$).
- Lt: Updated level at time t.

3. **Seasonality Component (St):**

$$S_t = \gamma \cdot \frac{y_t}{L_t} + (1 - \gamma) \cdot S_{t-m}$$

- St: Seasonal factor at time t.
- γ: Smoothing parameter for seasonality ($0 < \gamma \leq 1$).
- Lt: Updated level at time t.

Forecasting

Once the components (Lt, Tt, and St) are estimated, forecasts for h time steps ahead (yt+h) can be calculated using the following formula:

$$\hat{y}_{t+h} = (L_t \cdot T_t^h) \cdot S_{t+h-mk}$$

- Tt^h: Trend component extrapolated for h steps ahead.
- St+h−mk: Seasonal factor for the forecasted period, where k is the number of complete seasonal cycles between t and t+h.

Key Assumptions

1. **Multiplicative Seasonality**: Seasonal effects increase or decrease proportionally with the level.
2. **Stationary Trend and Seasonality**: The seasonal pattern repeats consistently over time.

Initialization

The initial values for L0, T0,and S0 must be computed before starting the recursive updates:

- L0: Average of the first m observations, adjusted for seasonality.

$$L_0 = \frac{1}{m} \sum_{t=1}^{m} \frac{y_t}{S_t}$$

- T0: Average change between consecutive levels over the first m periods.

$$T_0 = \frac{1}{m} \sum_{t=1}^{m-1} \frac{L_{t+1}}{L_t}$$

- S0: Seasonal factors for the first mmm periods, calculated as:

$$S_t = \frac{y_t}{L_0},\ t = 1, 2, ..., m$$

Smoothing Parameters

The parameters α, β, and γ are typically chosen to minimize the sum of squared errors (SSE) between the actual and fitted values:

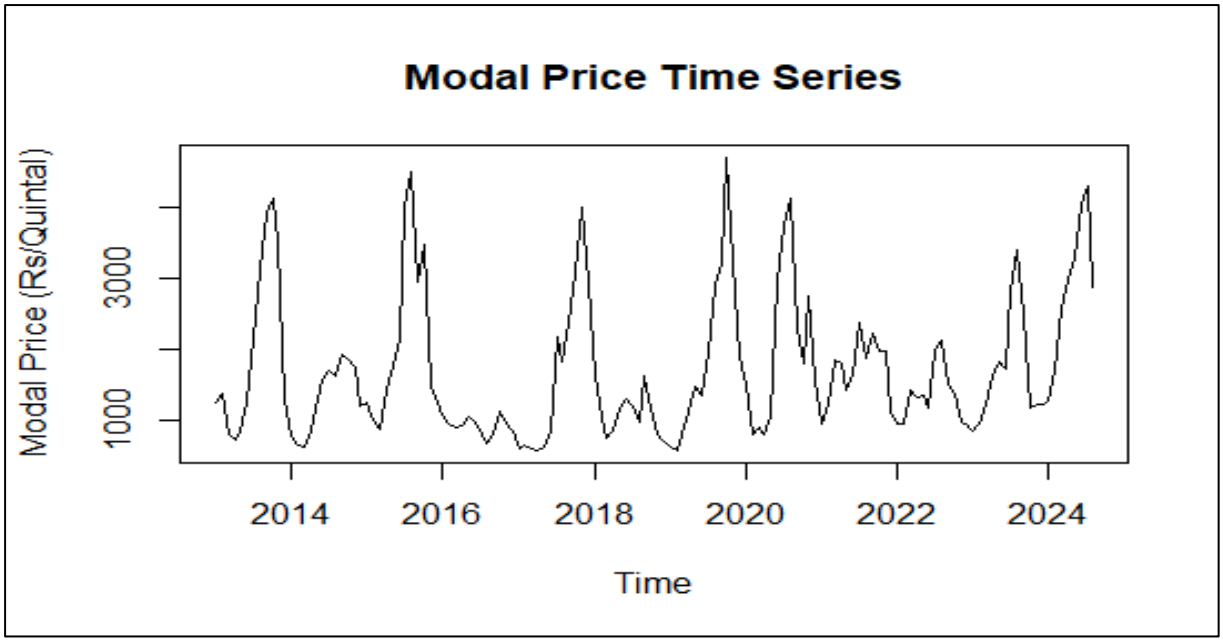
$$SSE = \sum_{t=1}^{n} (y_t - \hat{y}_t)^2$$

Advantages

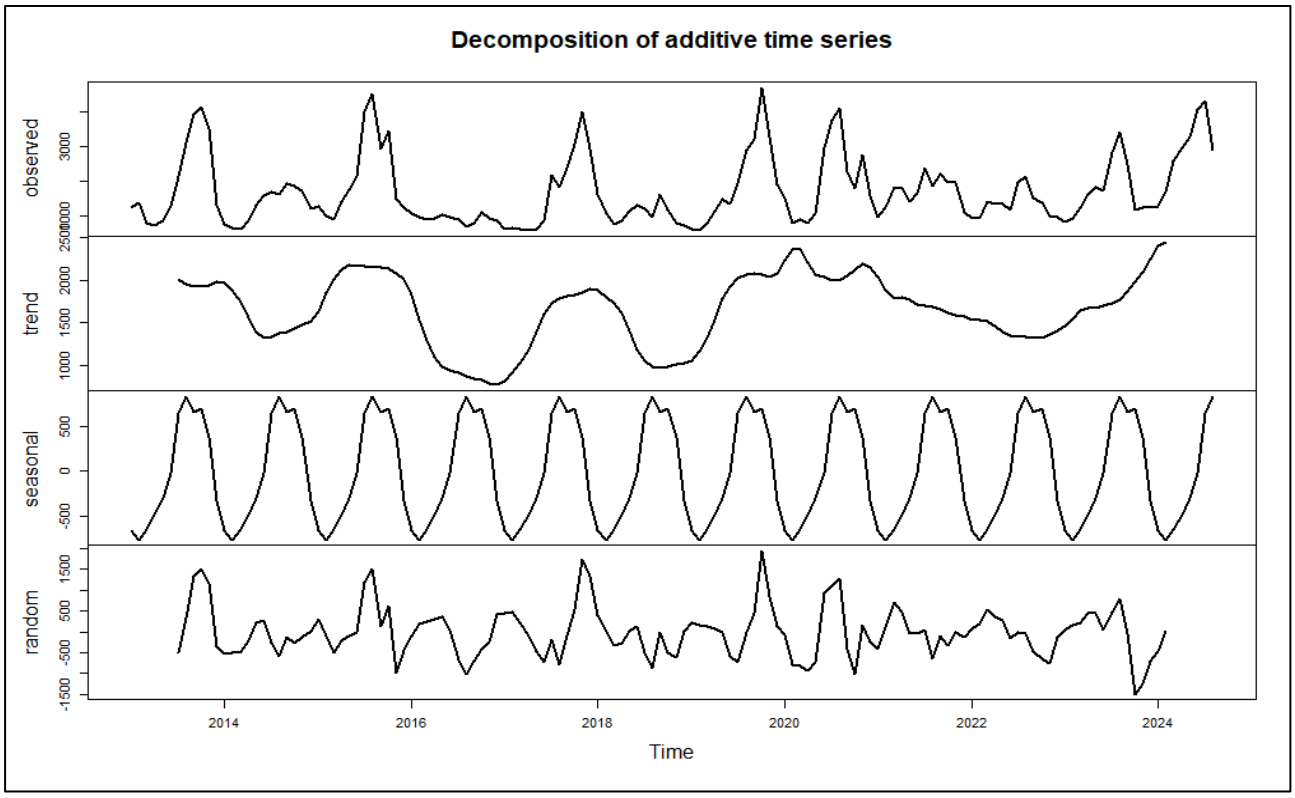
1. Handles both trend and seasonality effectively.
2. Flexible due to adjustable smoothing parameters.
3. Provides a simple yet powerful forecasting tool for seasonal data.

Limitations

1. Assumes constant periodicity of seasonality.
2. Requires a significant amount of historical data for accurate initialization and modeling.
3. Not suitable for non-stationary seasonality or irregular patterns.

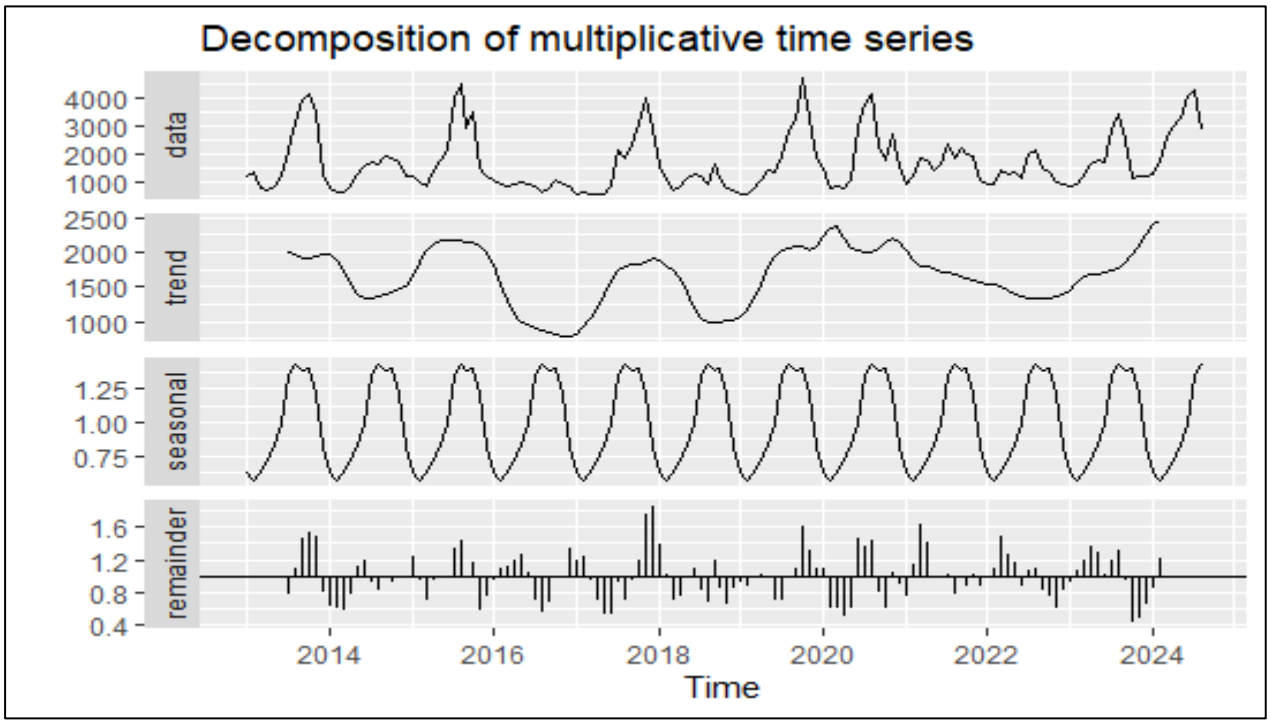


The plot shows the **time series of modal prices** (Rs/Quintal) from 2013 to 2024, highlighting strong **seasonality** with recurring peaks and troughs every 1-2 years, driven by factors like crop cycles and demand-supply dynamics. Prices range from **1000 to 3000 Rs/Quintal**, indicating significant **volatility**, with major spikes in **2016, 2020, and 2024**, suggesting market disruptions or increased demand. The predictable seasonal patterns can aid in **forecasting** and planning, while the sharp fluctuations emphasize the need for better risk management and market interventions.



The additive decomposition breaks the time series into **trend**, **seasonal**, and **random** components:

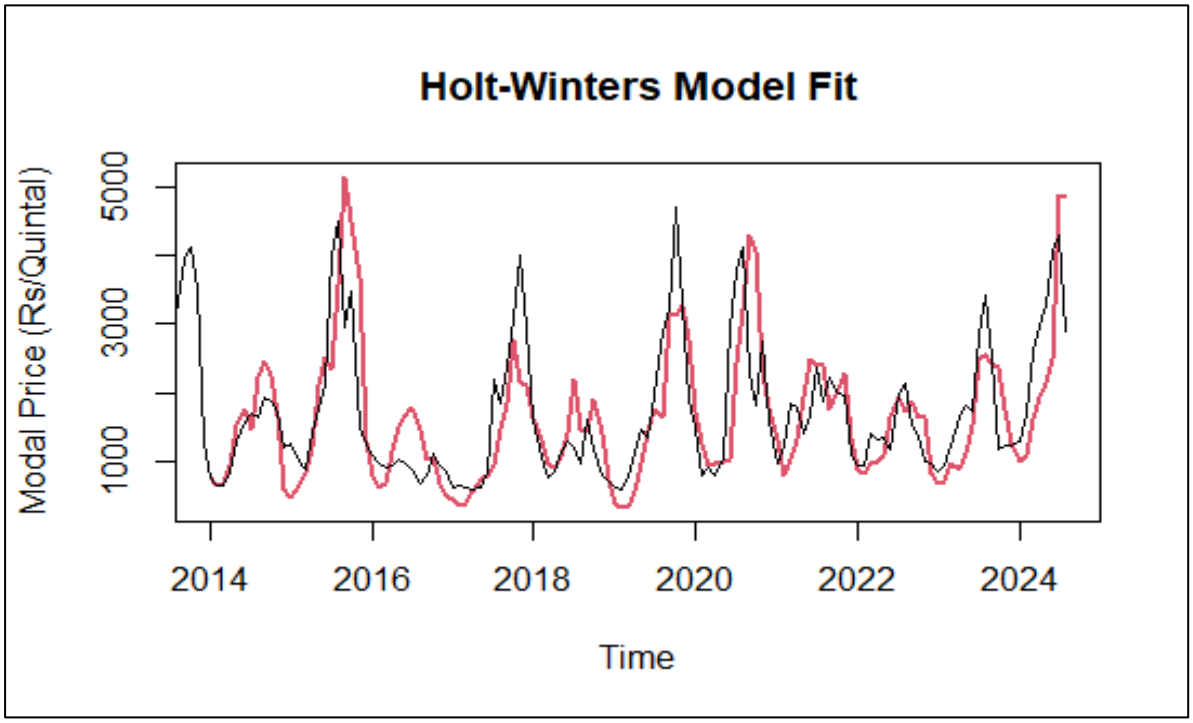
- The **trend** shows a rise until 2016, a dip between 2017–2020, and recovery towards 2024.
- The **seasonal** component highlights consistent, recurring patterns driven by predictable factors like crop cycles.
- The **random** component captures irregular variations caused by external factors like market shocks or weather disruptions.



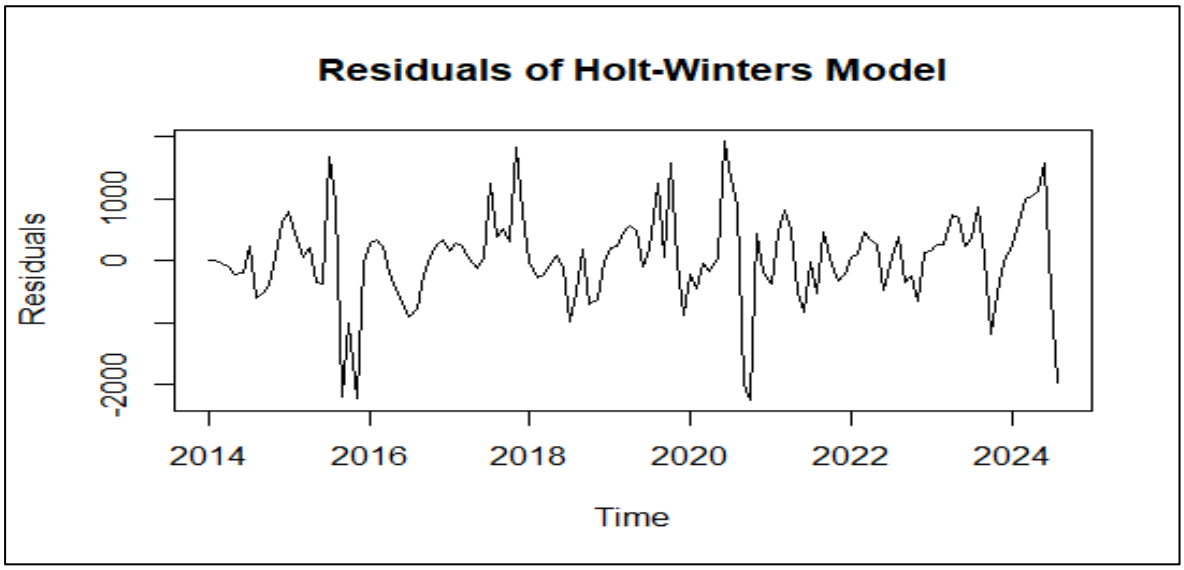
The multiplicative decomposition reveals:

- The **trend** follows a similar pattern, with a decline from 2016 to 2020 and a rise post-2021.
- The **seasonal** component indicates that seasonal effects scale proportionally with price levels, amplifying during peaks.
- The **remainder** highlights proportional irregularities, reflecting anomalies during extreme events.

Model Fitting

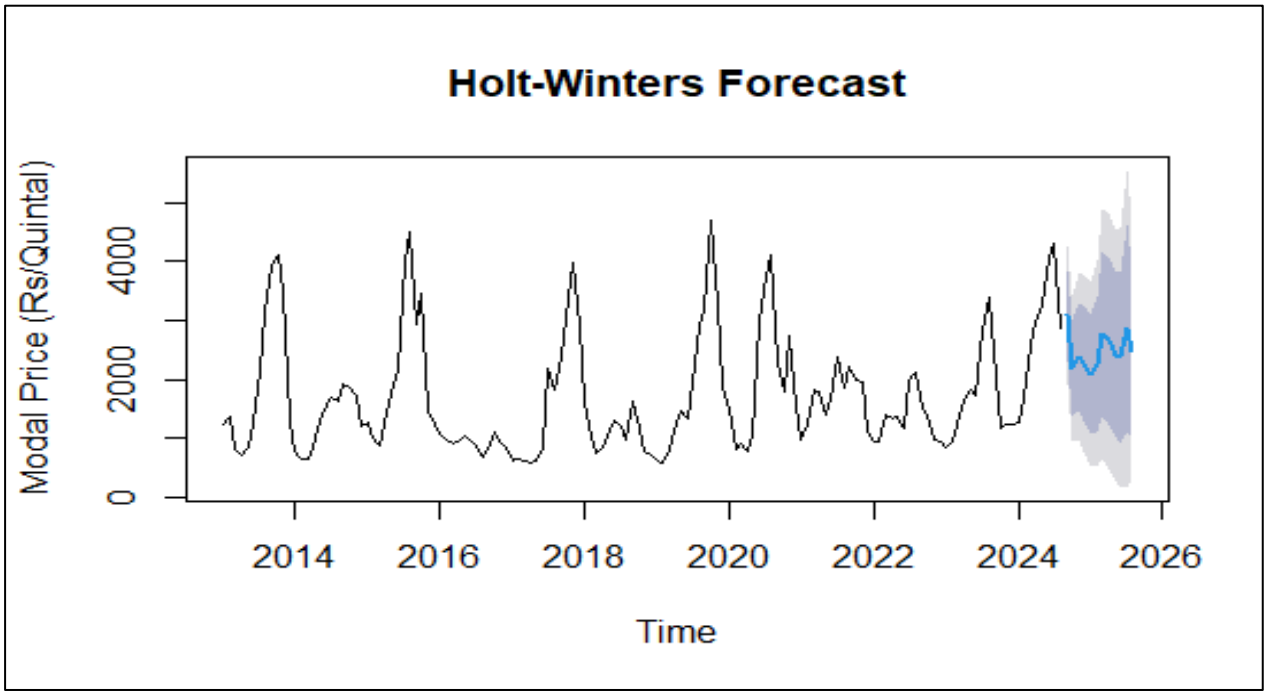


The Holt-Winters model demonstrates strong performance in capturing the seasonal patterns and overall trend in modal onion prices from 2013 to 2024. It closely follows observed peaks and troughs, indicating high accuracy and reliability in forecasting future prices. The model effectively accounts for significant price volatility during specific periods, showcasing its robustness and adaptability in handling seasonal and market-driven fluctuations. This makes it a suitable tool for predicting onion price dynamics.



The **Box-Ljung test** indicates a test statistic (X-squared) of 36.842 with a p-value of **0.0002369**. Since the p-value is much smaller than 0.05, we reject the null hypothesis that the residuals are uncorrelated. This suggests that the **residuals** of the Holt-Winters model exhibit autocorrelation, indicating the model may not fully capture all patterns in the data.

Forecasted Model



The Holt-Winters forecast accurately captures historical seasonality and trends, with reliable short-term predictions (2024-2026). However, widening confidence intervals highlight increasing uncertainty, emphasizing the model's effectiveness but limited long-term precision.

Coclusions:

1)During COVID, onion arrivals increased compared to pre-COVID but were lower than post-COVID levels , reflecting supply chain recovery. Prices significantly rose during COVID ($p < 2.2e-16$) and remained slightly higher post-COVID ($p = 1.42e-12$), indicating lasting pandemic effects on supply and pricing.

2)Onion prices significantly increased from pre-COVID to during COVID and further rose post-COVID, reflecting sustained price effects. While there was no significant change in arrivals between pre-COVID and during COVID, arrivals during COVID were significantly higher than post-COVID, indicating supply chain adjustments post-pandemic.(pune District)

3) The SARIMA model forecast indicates a slight increase in onion prices for 2025–2026, continuing the observed seasonal trends from historical data. The forecast captures seasonal patterns effectively, with confidence intervals showing increasing uncertainty over time.

4) The Holt-Winters model effectively captures the seasonality and trends in onion prices, providing reliable short-term forecasts (2025–2026).

5) The SARIMA (2,1,0)(1,1,1) model is the better choice. It offers superior training fit, can handle complex seasonality, and is likely to provide better forecast accuracy than the Holt-Winters model, which struggles with capturing the complexities of the dataset.

References:

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4) https://www.google.com/